Spectral Shrunk Clustering: Reproduction and Independent Refinement Studies

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Abstract

Spectral Shrunk Clustering (SSC) refines a spectral representation through a convex fidelity–fusion objective. We separate paper reconstruction, numerical optimization and clustering quality. Joint ACC/NMI matching and declared source/subset sensitivities preserve distinct statistical and identity qualifications. Certified shrinkage has small, mixed net effects at gamma 1; normalization and graph interventions expose different limitations. An oracle graph-completion diagnostic passes on ORL. Three separately registered convex refinements test dispersion protection, tangent-constrained motion and frozen relative-neighbor corrections; their numerical qualification and mechanism gates remain distinct from clustering performance. Source and implementation audits retain failures and adapted methods explicitly. Numerical agreement does not identify the authors' historical datasets.

1 What SSC optimizes

Spectral clustering (SC) represents samples using graph-Laplacian eigenvectors and then applies K-means. Chang et al. [2015] argue that directly clustering the initial spectral coordinates can be unreliable, and propose learning *shrunk patterns* that preserve their anchors while contracting neighboring representations. Their contribution is a convex representation-refinement formulation, relaxation of an orthogonality constraint in this stage, and an iterative algorithm with clustering experiments. SSC here means Spectral Shrunk Clustering, rather than Sparse Subspace Clustering.

The input-space affinity A determines $L_A = I - D_A^{-1/2} A D_A^{-1/2}$. The c smallest eigenvectors form $F \in \mathbb{R}^{n\times c}$. A second affinity W is constructed from F and controls refinement (Figure 1). With each undirected edge stored once in incidence matrix B, Equation 8 is

$$P(G) = \sum_i \|g_i - f_i\|_2 + 2\gamma \sum_e w_e \|(BG)_e\|_2. \tag{1}$$

The factor 2 implements the paper's ordered-pair sum. Its stationary IRLS system uses $S + 2\gamma\widetilde{L}$ (E8), whereas printed Equation 13 uses $S + \gamma\widetilde{L}$ (E13). Thus E13 at nominal gamma targets E8 at effective gamma/2. We retain both tracks, rather than calling unequal targets a solver discrepancy.

For fixed F, W, nonnegative weights make (1) convex; unsquared fidelity does not guarantee uniqueness. Graph construction, eigenspace selection and final K-means are separate operations. A globally optimal G therefore need not improve ACC or NMI.

2 Published results and the reproduction protocol

We verified all 288 reported means and SD values in the paper's Tables 1–2: nine methods, eight benchmarks and two metrics. The paper states $k = 5, 50$ random initializations, a five-value parameter

Preprint.

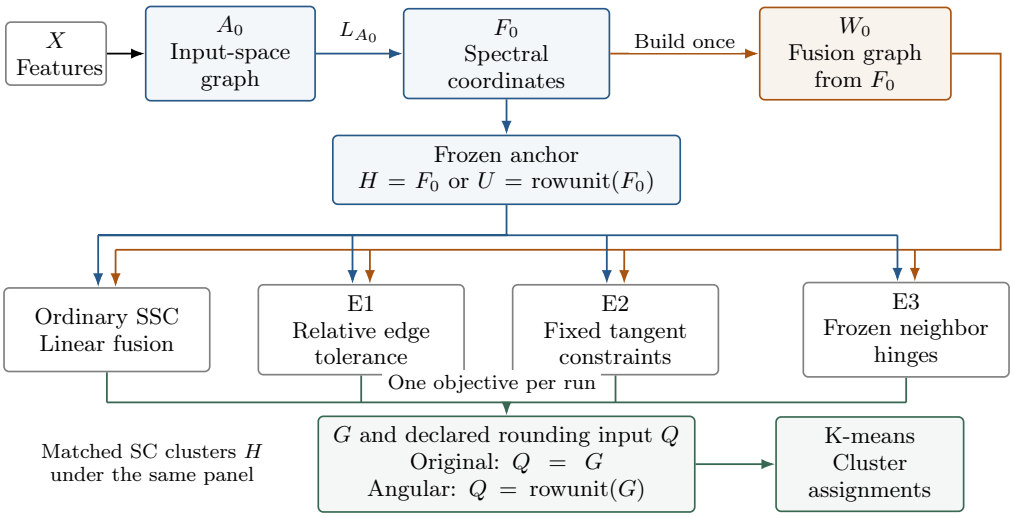


Figure 1: The input graph A_0 determines F_0; the fusion graph W_0 is built once from F_0. Each refinement uses frozen inputs and produces its own G. The original and common angular panels declare their respective rounding matrices Q. The three adaptations are alternatives.

grid and native-objective selection. It does not fully resolve the SD hierarchy, source indices, Gaussian width or graph symmetrization. Our choices are union kNN, median positive directed-neighbor squared-distance width, deterministic eigensolver initialization, and explicit one-start Lloyd rounding. ACC uses optimal label assignment and NMI uses geometric entropy normalization.

The paper's SSC ACC gains do not imply superiority on every metric: YaleB SC/SSC have ACC 34.8/49.8 but NMI 55.6/46.8. We therefore minimize joint discrepancy in percentage points,

$$D_{\max} = \max(|\overline{\text{ACC}} - \text{ACC}_p|, |\overline{\text{NMI}} - \text{NMI}_p|), \qquad D_{\text{mean}} = \tfrac{1}{2}(d_{\text{ACC}} + d_{\text{NMI}}).$$

We rank by $D_{\max}$, then D_{mean}, with a 10^{-12} tolerance and frozen compatibility tie breaks. Published-grid gamma values are not penalized for differing from 1. Both metrics and SD come from one configuration and its complete 50 seeds. The single native-best-of-50 assignment has no across-run SD; existing independent batches are a separate repeatability endpoint. Raw LDA inertia across learned subspaces is not treated as a common selection objective.

Data	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$	Fit
AR	51.30 ± 1.50	51.21 ± 2.00	73.20 ± 1.40	72.94 ± 1.05	0.26	T
JAFFE	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45	T
ORL	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37	I
UMIST	71.10 ± 1.80	69.97 ± 2.27	84.10 ± 1.50	85.17 ± 1.00	1.13	T
BinAlpha	49.40 ± 1.30	49.54 ± 2.71	64.10 ± 1.40	63.46 ± 1.04	0.64	I
MSRA50†	63.20 ± 1.10	49.25 ± 4.38	72.20 ± 1.40	61.50 ± 3.89	13.95	T
YaleB	49.80 ± 1.60	44.24 ± 0.82	46.80 ± 1.30	52.61 ± 0.36	5.81	T
USPS	75.50 ± 1.90	66.26 ± 1.70	79.80 ± 1.60	80.86 ± 1.20	9.24	I

Table 1: Closest evaluated SSC reconstructions under joint $D_{\max}/D_{\text{mean}}$ matching. This is retrospective numerical selection using labels and paper targets; historical subset identity remains unverified. Means $\pm$ sample SD use one configuration and all 50 seeds. Compatible-pool winners are shown where available; † marks a separately selected adaptation. I: iterate stopping without a gap certificate; T: iteration cap; U: other uncertified state. Exact candidate IDs, counts, signed differences, settings and both pools appear in the appendix. Historical selections are retained separately.

2.1 Dataset-subset uncertainty and retrospective numerical matching

Original sample indices were unavailable. Final source/subset matches and locked-selection rounding are reported separately from the historical and method-comparison endpoints. **Such matching is**

retrospective, not label-free selection or an unbiased superiority test. Exact manifests support replay, not historical identity. Changed membership rebuilds A, F, W, G; source, resize, normalization and eligibility pools remain distinct (Appendix C).

The evaluated design obtained complete SSC configurations on 2482 unique inputs, yielding 74436 input/configuration pairs. Of these, 2480 inputs completed all 30 SSC settings and 2480 completed the 30-SSC plus three matched-SC design. Aliases are provenance entries, and diagnostic parents are excluded from equivalent paper reconstruction. Additional rounding on locked selections measures conditional stability; closer numerical matching does not identify historical data.

3 What discrepancies can be explained

Numerics. For $c_e = 2\gamma w_e$, feasible q with $\|q_e\| \leq c_e$ and $\|(B^T q)_i\| \leq 1$ gives lower bound $D(q) = \langle B^T q, F\rangle$. We retain the initial feasible bound in preconditioned PDHG [Chambolle and Pock, 2011]; certification requires relative gap at most 10^{-5} and capacity violations at most 10^{-8}. On one frozen current USPS F/W, the gamma-1 continuation certifies its saved primal at gap 9.9996×10^{-6}, yet ACC/NMI change by only $-0.033/-0.086$ points against same-F SC. Gamma 1000 remains uncertified at gap 0.342; its collapsed saved point does not prove optimal collapse. Failed IRLS/dual-flow attempts and missing historical arrays remain separate limitations.

Implementation and sources. The pinned ClustPy implementation of LDA-Kmeans [Ding and Li, 2007] failed every LDA update in all 50 ORL and JAFFE fits; these are implementation-specific PCA–K-means degeneracies. The projection actually clustered and the source-returned projection are recorded separately. Existing-project LDA limits 100/300 give identical per-seed ACC/NMI on all five tested sources. The separately named ridge LDA-Kmeans→SC adaptation improves both means over raw/PCA_{p_0} SC only on YaleB; it does not recover historical LDA-SC. Paired SLPS [Hou et al., 2013] $k = 5/7$ effects change with source and alpha. Native 16-by-16 EBSA MSRA retains all 1,799 rows and duplicates. Exact pins, adaptation formulas and controls are in Appendix C.

The baseline comparison contains 98 complete 50-seed SLPS configurations, 30 LDA configurations/control endpoints and 62 other configurations. The appendix retains numerical failures, iteration-limit outcomes and update diagnostics. Native 16-by-16 MSRA and ridge LDA-to-SC are explicitly named adaptations; exact historical LDA-SC remains unresolved.

4 Graph diagnostics

G1 compares SC/SSC on the same fixed representation. Certified gamma-1 shrinkage changes mean ACC/NMI by $+0.155/+0.014$ points on ORL and $-0.270/-0.276$ on YaleB, despite active movement. A and W have different roles: cross-class edges carry 4.6% of ORL W mass but 70.2% of YaleB W mass. Class-induced components, including isolates, distinguish within-class fragmentation from whole-graph connectivity. YaleB's W edge-length AUROC is 0.380, weakening the general premise that erroneous edges are longer; these associations alone do not establish their causal effect.

G2 separates anchor normalization from the fusion graph at matched mass. Row-normalized U alone raises SC ACC by 1.850/4.016 points on ORL/YaleB; angular-graph shrinkage at gamma 1 adds only 0.170/0.314 above U-SC. A stronger representation baseline therefore matters more than a comparison only with F_0-SC.

The fixed G3 labelled intervention diagnoses whether correcting graph relationships can help. ORL passes; YaleB remains numerically unqualified. This is a mechanism experiment, not a label-free method result.

5 Three independent refinement studies

With $H = F_0$ or $U = \text{rowunit}(F_0)$ and W_0 frozen, E1 replaces each edge norm by $(\|(BG)_e\| - \eta\|(BH)_e\|)_+$, with $\eta = .25$. Its optimum retains at least η of global geometric-median dispersion. This does not protect individual classes, guarantee improved ACC/NMI or change the threshold for leaving the anchor. Interpolation and uniform origin scaling distinguish useful refinement from simple damage avoidance.

E2 constrains each displacement at nonzero u_i to its fixed tangent plane and radius $\tan 15^\circ$; zero anchors stay fixed. It is promoted only if a predefined ordinary-SSC radial-neutralization contrast improves ACC with a positive adjusted lower bound, preserves mean NMI, has at least 10% inward energy and changes direction by at least 1° on either development dataset. An isotropic ball of radius $\sin 15^\circ$ is a required control.

E3 adds unit-gradient affine hinges from two-view agreement and contradictory W_0 neighbors, with $\mu = 2$. All three sample roles come from the common valid pool. Triplets are frozen before labels are used for the development semantic audit; promotion requires sufficient coverage, at least 60% correct relations and at most 10% reversed relations. Triplet-only correction and three matched random-direction controls separate geometric information from generic extra force.

Fixed primary effects. Differences are percentage points against matched ordinary SSC; C means both saved outputs are certified. A gated method has no formal benchmark score. All historical identities remain unverified; MSRA denotes the adapted native-16 proxy.

Data	E1 orig. ΔACC	ΔNMI	E1 angular ΔACC	ΔNMI
AR	0.045 C	0.042	-0.052 C	-0.022
JAFFE	0.000 C	0.000	0.000 C	0.000
ORL	-0.180 C	0.206	0.010 C	0.008
UMIST	0.638 C	0.113	0.004 C	-0.003
BinAlpha	0.415 C	0.292	0.249 C	0.101
MSRA50	-0.010 U	0.066	-0.071 C	-0.016
YaleB	0.030 C	0.002	-0.214 C	-0.108
USPS	0.053 C	0.153	0.012 C	0.108

E2: development gate not passed.

E3: development gate not passed.

Eight-input qualified paired macro: E1 original: unavailable (7/8); E1 angular: -0.008 ACC / 0.008 NMI; E2 angular: unavailable (0/8); E3 angular: unavailable (0/8).

The two panels are evaluated separately: original geometry rounds G, while every angular arm rounds $\mathrm{rowunit}(G)$. The label-free operating point is $2\gamma_{\mathrm{ref}}$, defined from the anchor's maximum node fusion force. The same complete 50 rounding seeds produce ACC/NMI means and sample SD. Conditional paired intervals use 10,000 resamples. Development paths and available certified movement matches cannot replace the primary point. An eight-input macro is unavailable until all eight paired outputs qualify. E1 also changes angular ACC/NMI by $-0.040/+0.013$ points against interpolation, so the dispersion guarantee has not yielded a practical advantage over that simple control. E2 and E3 are not promoted: the radial intervention is too small and triplet semantics are insufficient. Complete controls, failures and derivations are retained in the appendix and scalar release.

6 Conclusion

SSC gives a clear convex refinement target, while clustering quality depends on representation, graph, solver and rounding choices. Our contribution is reproducible source/subset sensitivity, numerical audits and independently gated refinements. E1 preserves dispersion without a practical mean clustering gain here; the development evidence does not support promoting E2 or E3. Exact historical LDA-SC and several dataset identities remain unresolved. Code, statistical records and reproducible commands are available in the public repository.

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A Historical measurements and endpoint separation

The original scalar measurements and ACC-only selections remain distinct historical endpoints. Joint selections use complete seed groups and preserve their original measurements. Missing historical representations limit optimization re-audits on their exact arrays. The historical 72-cell identity/status matrix remains separate from newly measured candidates and adaptations. The original ACC-only closest-display and fixed-reference tables remain available in the committed historical release. The current joint 72-cell table below supersedes their display selection without rewriting any trial.

B Results, statistical endpoints and coverage

B.1 Jointly reselected historical mean endpoint

Table 2 reports the complete 72-cell inventory. Values are percent. Every measured pair uses the same configuration and 50 batch-zero seeds; $\pm$ denotes sample SD. Missing cells retain their identity or method restriction. Historical scalar evidence is separate from new recoverable artifacts. The former ACC-only and fixed-reference tables remain in the committed historical release.

Table 2: Joint closest historical mean/sample-SD endpoint; unavailable cells are not filled with adapted measurements.

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR	K-means	36.30 ± 1.40	–	68.70 ± 3.00	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SC	41.60 ± 2.10	–	71.30 ± 2.60	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–K-means	39.80 ± 1.80	–	69.40 ± 2.80	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–SC	43.20 ± 1.70	–	70.30 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–K-means	40.40 ± 1.50	–	69.90 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–SC	44.50 ± 1.30	–	70.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LLC	48.70 ± 1.60	–	71.20 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SLPS	49.20 ± 1.40	–	72.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SSC	51.30 ± 1.50	–	73.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	K-means	75.60 ± 1.80	71.97 ± 7.55	79.40 ± 0.80	81.97 ± 3.70	3.63
JAFFE; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 3.10$.						
JAFFE	SC	76.10 ± 1.60	68.53 ± 11.15	80.20 ± 0.90	83.02 ± 5.72	7.57
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 5.20$.						
JAFFE	PCA–K-means	75.80 ± 1.50	73.20 ± 5.58	79.80 ± 0.80	82.95 ± 3.67	3.15
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.88$.						
JAFFE	PCA–SC	76.90 ± 1.80	72.30 ± 0.00	81.50 ± 1.30	82.86 ± 0.00	4.60
JAFFE; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.98$.						
JAFFE	LDA–K-means	76.50 ± 1.70	64.44 ± 9.19	82.10 ± 1.40	78.60 ± 5.08	12.06
JAFFE; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 7.78$.						
JAFFE	LDA–SC	77.40 ± 1.90	–	81.90 ± 0.90	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	LLC	78.60 ± 1.50	91.55 ± 0.00	82.50 ± 1.70	92.75 ± 0.00	12.95
JAFFE; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 11.60$.						
JAFFE	SLPS	79.50 ± 2.10	80.86 ± 5.77	83.10 ± 2.10	86.66 ± 3.34	3.56
JAFFE; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.46$.						
JAFFE	SSC	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45
JAFFE; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $D_{\text{mean}} = 3.87$.						
ORL	K-means	60.50 ± 1.80	61.46 ± 2.85	80.30 ± 1.80	82.76 ± 1.17	2.46
ORL_att92; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 1.71$.						
ORL	SC	72.70 ± 2.30	72.73 ± 3.15	85.80 ± 1.90	87.49 ± 1.22	1.69
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 0.86$.						
ORL	PCA–K-means	64.50 ± 2.30	63.08 ± 2.26	80.60 ± 1.60	82.34 ± 0.90	1.74
ORL_att92; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 1.58$.						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
ORL	PCA–SC	67.80 ± 2.10	67.57 ± 2.92	86.30 ± 1.40	85.81 ± 1.27	0.49
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.36$.						
ORL	LDA–K-means	65.60 ± 2.60	56.31 ± 4.14	81.10 ± 2.10	81.00 ± 1.65	9.29
ORL_att92; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\mathrm{mean}} = 4.70$.						
ORL	LDA–SC	68.30 ± 2.40	–	86.80 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
ORL	LLC	71.50 ± 2.20	81.07 ± 0.61	84.90 ± 1.50	89.92 ± 0.27	9.57
ORL_att92; Y0, YU_SHI; NOT_APPLICABLE; $D_{\mathrm{mean}} = 7.29$.						
ORL	SLPS	74.20 ± 1.80	73.66 ± 2.13	87.20 ± 1.80	87.71 ± 0.91	0.54
ORL_att92; Y0, one_shot, G; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.53$.						
ORL	SSC	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37
ORL_att92; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $D_{\mathrm{mean}} = 1.24$.						
UMIST	K-means	41.30 ± 1.60	–	64.40 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SC	52.20 ± 1.40	–	72.10 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–K-means	48.80 ± 1.70	–	68.20 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–SC	54.10 ± 1.90	–	72.90 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–K-means	49.70 ± 1.80	–	68.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–SC	54.70 ± 1.50	–	74.10 ± 2.00	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LLC	63.30 ± 1.80	–	77.30 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SLPS	70.40 ± 1.60	–	82.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SSC	71.10 ± 1.80	–	84.10 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	K-means	41.70 ± 1.10	–	58.60 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SC	43.60 ± 1.50	–	59.70 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–K-means	42.40 ± 1.50	–	59.10 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–SC	44.30 ± 1.80	–	60.60 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–K-means	42.90 ± 1.70	–	59.80 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–SC	45.10 ± 1.40	–	61.30 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LLC	40.70 ± 1.80	–	61.40 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SLPS	48.50 ± 1.90	–	63.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SSC	49.40 ± 1.30	–	64.10 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	K-means	46.20 ± 1.70	–	56.70 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SC	52.30 ± 1.80	–	70.00 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–K-means	56.00 ± 1.90	–	60.30 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–SC	54.90 ± 1.60	–	72.40 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–K-means	56.40 ± 1.80	–	61.10 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–SC	55.10 ± 1.70	–	73.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LLC	48.10 ± 1.40	–	66.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SLPS	60.30 ± 1.30	–	69.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SSC	63.20 ± 1.10	–	72.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
YaleB	K-means	14.40 ± 1.50	9.74 ± 0.46	17.30 ± 1.50	13.25 ± 0.59	4.66
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.35$.						
YaleB	SC	34.80 ± 1.40	37.39 ± 2.22	55.60 ± 1.60	52.96 ± 1.00	2.64
YaleB_dengcai32; Y3, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.61$.						
YaleB	PCA–K-means	24.90 ± 1.80	9.85 ± 0.55	26.70 ± 1.80	13.32 ± 0.66	15.05
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 14.22$.						
YaleB	PCA–SC	36.80 ± 2.00	30.92 ± 0.54	38.60 ± 1.50	38.86 ± 0.95	5.88
YaleB_dengcai32; Y1, G; NOT_APPLICABLE; $D_{\text{mean}} = 3.07$.						
YaleB	LDA–K-means	26.10 ± 1.80	9.80 ± 0.85	29.40 ± 1.60	13.79 ± 1.15	16.30
YaleB_dengcai32; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 15.96$.						
YaleB	LDA–SC	38.20 ± 1.60	–	39.90 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
YaleB	LLC	38.20 ± 1.50	32.35 ± 0.05	34.10 ± 1.30	39.68 ± 0.11	5.85
YaleB_dengcai32; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 5.72$.						
YaleB	SLPS	47.30 ± 1.70	13.52 ± 0.78	41.40 ± 1.60	19.44 ± 0.69	33.78
YaleB_dengcai32; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 27.87$.						
YaleB	SSC	49.80 ± 1.60	42.64 ± 0.61	46.80 ± 1.30	52.19 ± 0.31	7.16
YaleB_dengcai32; Y3, E8, $\gamma = 1e+06$, G; MAX_ITER_REACHED; $D_{\text{mean}} = 6.27$.						
USPS	K-means	65.40 ± 1.70	66.06 ± 2.17	67.30 ± 1.80	61.04 ± 0.87	6.26
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 3.46$.						
USPS	SC	64.30 ± 1.40	61.74 ± 8.09	69.50 ± 1.60	74.94 ± 4.49	5.44
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 4.00$.						
USPS	PCA–K-means	69.40 ± 1.80	65.25 ± 2.85	73.10 ± 1.90	61.10 ± 0.95	12.00
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 8.08$.						
USPS	PCA–SC	69.10 ± 1.50	65.53 ± 3.26	74.20 ± 1.80	78.92 ± 2.12	4.72
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.15$.						
USPS	LDA–K-means	70.10 ± 1.30	58.61 ± 6.33	75.10 ± 1.60	60.29 ± 4.21	14.81
USPS; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 13.15$.						
USPS	LDA–SC	70.40 ± 1.50	–	75.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
USPS	LLC	63.90 ± 1.70	65.27 ± 0.01	67.50 ± 1.50	72.76 ± 0.01	5.26
USPS; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 3.32$.						
USPS	SLPS	71.40 ± 1.60	70.81 ± 8.33	76.50 ± 1.80	76.30 ± 2.47	0.59
USPS; Y0, one_shot, R; NOT_APPLICABLE; $D_{\text{mean}} = 0.39$.						
USPS	SSC	75.50 ± 1.90	73.87 ± 7.53	79.80 ± 1.60	82.84 ± 3.96	3.04
USPS; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $D_{\text{mean}} = 2.34$.						

B.2 Native-objective-selected endpoint

Within one fixed G, native K-means SSE selects one of 50 assignments. This is a single assignment with no across-run SD. It does not compete with the mean endpoint. Full method-specific endpoint records are in `results/submission_refresh/endpoint_comparison.csv`; LDA cross-subspace native selection is unavailable.

SSC data	ACC	NMI	$D_{\max}$	D_{mean}	Seed	Fit
AR	–	–	–	–	–	–
JAFFE	95.31	94.09	14.11	11.95	13	T
ORL	75.00	88.60	1.00	0.50	40	I
UMIST	–	–	–	–	–	–
BinAlpha	–	–	–	–	–	–
MSRA50	–	–	–	–	–	–
YaleB	43.33	54.15	7.35	6.91	45	I
USPS	79.62	85.38	5.58	4.85	38	I

B.3 Separate historical repeatability batches

Existing multi-batch records are a separate endpoint. Each batch contains 50 disjoint rounding seeds on its fixed input. The following ranges describe variation of batch means and native-selected assignments, without treating a single native assignment as having a within-batch SD. They are not independent datasets, and are not new campaign measurements. All methods and exact configuration IDs are preserved in `historical_repeatability_batches.csv` and its summary/manifest. LDA source-native selection remains diagnostic only.

SSC setting	Batches	Batch ACC	Batch NMI	Native ACC	Native NMI
BinAlpha_candidate26x30 / E13 / gamma=1e+00 / R	10	42.21–43.23	56.50–57.12	44.62–47.82	57.53–59.66
BinAlpha_candidate26x30 / E13 / gamma=1e+00 / G	10	47.05–47.52	58.55–58.91	45.51–49.36	58.37–59.70
BinAlpha_candidate26x30 / E8 / gamma=1e+00 / R	10	40.54–41.55	55.87–56.61	43.59–45.90	57.04–59.18
BinAlpha_candidate26x30 / E8 / gamma=1e+00 / G	10	46.35–47.18	58.33–59.09	45.38–48.33	57.06–59.31
JAFFE / E13 / gamma=1e+00 / R	10	61.13–68.59	78.60–83.04	80.75–95.31	89.37–94.16
JAFFE / E13 / gamma=1e+00 / G	10	95.06–95.31	94.05–94.14	95.31–95.31	94.12–94.12
JAFFE / E8 / gamma=1e+00 / R	10	61.08–68.43	78.70–83.05	80.75–95.31	89.37–94.16
JAFFE / E8 / gamma=1e+00 / G	10	94.85–95.31	93.99–94.14	95.31–95.31	94.12–94.12
ORL_att92 / E13 / gamma=1e+00 / R	10	55.83–57.48	80.42–81.43	63.00–73.25	82.97–87.94
ORL_att92 / E13 / gamma=1e+00 / G	10	78.04–78.77	89.52–89.86	78.25–80.50	89.83–90.93
ORL_att92 / E8 / gamma=1e+00 / R	10	56.12–58.05	80.70–81.68	62.75–70.50	83.63–87.91
ORL_att92 / E8 / gamma=1e+00 / G	10	78.25–78.75	89.66–89.95	78.25–80.50	89.97–91.39
UMIST_precropped575 / E13 / gamma=1e+00 / R	10	54.34–57.24	75.69–77.00	56.70–67.48	76.40–82.39
UMIST_precropped575 / E13 / gamma=1e+00 / G	10	59.98–61.01	78.21–78.68	60.70–60.87	77.83–78.02
UMIST_precropped575 / E8 / gamma=1e+00 / R	10	54.49–56.87	74.95–76.16	54.61–66.61	76.62–81.01
UMIST_precropped575 / E8 / gamma=1e+00 / G	10	60.35–61.78	78.35–78.90	60.35–61.04	77.92–78.52
USPS / E13 / gamma=1e+00 / R	10	59.71–62.17	74.34–75.71	65.47–65.59	80.97–81.19
USPS / E13 / gamma=1e+00 / G	10	64.89–66.26	79.94–80.86	65.48–65.50	80.99–81.01
USPS / E8 / gamma=1e+00 / R	10	59.56–61.93	74.38–75.53	65.63–65.63	81.17–81.24
USPS / E8 / gamma=1e+00 / G	10	64.76–65.94	79.77–80.89	65.62–65.63	81.17–81.19
YaleB_cropped2414 / E13 / gamma=1e+00 / R	10	17.96–18.20	26.41–26.67	19.51–21.29	26.78–27.84
YaleB_cropped2414 / E13 / gamma=1e+00 / G	10	21.59–21.83	27.49–27.75	21.71–21.96	27.41–28.23
YaleB_cropped2414 / E8 / gamma=1e+00 / R	10	17.46–17.89	26.05–26.34	18.77–20.34	25.94–28.07
YaleB_cropped2414 / E8 / gamma=1e+00 / G	10	21.40–21.56	27.24–27.58	21.25–22.08	27.07–28.19
YaleB_dengcai32 / E13 / gamma=1e+00 / R	10	19.80–20.29	29.21–29.63	21.46–23.28	28.80–31.77
YaleB_dengcai32 / E13 / gamma=1e+00 / G	10	25.32–25.55	31.13–31.47	24.98–25.52	31.31–31.96
YaleB_dengcai32 / E8 / gamma=1e+00 / R	10	19.50–19.88	29.09–29.32	20.59–23.16	29.52–30.98
YaleB_dengcai32 / E8 / gamma=1e+00 / G	10	25.31–25.46	31.24–31.43	25.02–25.10	31.54–31.65

B.4 Dataset-subset matching: both endpoints and eligibility pools

The following rows use complete evaluated configurations only. Paper values are fixed SSC targets. R/V/G denote random, vanilla K-means++ and sklearn greedy K-means++; each is one Lloyd start. Specification-compatible dimensions/counts do not establish historical pixel identity. Adapted rows retain explicit source/resolution/preparation differences. Diagnostic parents cannot win either pool.

Mean/sample-SD over 50 fixed-G roundings

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR / compatible	AR_common_mask0092	51.30 ± 1.50	51.21 ± 2.00	73.20 ± 1.40	72.94 ± 1.05	0.26

$n = 840$; Y0, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -0.09$ pp; $\Delta_{\mathrm{NMI}} = -0.26$ pp; $D_{\mathrm{mean}} = 0.18$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.50$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.35$ pp.

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
JAFFE / compatible	JAFFE	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45
$n = 213$; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -6.45$ pp; $\Delta_{\mathrm{NMI}} = 1.30$ pp; $D_{\mathrm{mean}} = 3.87$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = 6.77$ pp; $\Delta_{\mathrm{SD,NMI}} = 2.72$ pp.						
ORL / compatible	ORL_att92	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37
$n = 400$; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.37$ pp; $\Delta_{\mathrm{NMI}} = 1.11$ pp; $D_{\mathrm{mean}} = 1.24$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.96$ pp; $\Delta_{\mathrm{SD,NMI}} = -1.27$ pp.						
UMIST / compatible	UMIST_LZUFE574_stratified_seed079	71.10 ± 1.80	69.97 ± 2.27	84.10 ± 1.50	85.17 ± 1.00	1.13
$n = 564$; Y0, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -1.13$ pp; $\Delta_{\mathrm{NMI}} = 1.07$ pp; $D_{\mathrm{mean}} = 1.10$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.47$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.50$ pp.						
BinAlpha / adapted	BinAlpha_first30	49.40 ± 1.30	50.13 ± 1.39	64.10 ± 1.40	61.94 ± 1.07	2.16
$n = 780$; Y0, E8, $\gamma = 1$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = 0.73$ pp; $\Delta_{\mathrm{NMI}} = -2.16$ pp; $D_{\mathrm{mean}} = 1.44$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 0.09$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.33$ pp.						
BinAlpha / compatible	BinAlpha_random_seed057	49.40 ± 1.30	49.54 ± 2.71	64.10 ± 1.40	63.46 ± 1.04	0.64
$n = 780$; Y0, E8, $\gamma = 1e+06$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = 0.14$ pp; $\Delta_{\mathrm{NMI}} = -0.64$ pp; $D_{\mathrm{mean}} = 0.39$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 1.41$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.36$ pp.						
MSRA50 / adapted	MSRA25_EBSA_16x16	63.20 ± 1.10	49.25 ± 4.38	72.20 ± 1.40	61.50 ± 3.89	13.95
$n = 1799$; Y0, E13, $\gamma = 1e+06$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -13.95$ pp; $\Delta_{\mathrm{NMI}} = -10.70$ pp; $D_{\mathrm{mean}} = 12.32$; UNVERIFIED. $\Delta_{\mathrm{SD,ACC}} = 3.28$ pp; $\Delta_{\mathrm{SD,NMI}} = 2.49$ pp.						
YaleB / compatible	YaleB_cropped2414_nearest_Y3	49.80 ± 1.60	44.24 ± 0.82	46.80 ± 1.30	52.61 ± 0.36	5.81
$n = 2414$; Y3, E13, $\gamma = 1000$, G; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -5.56$ pp; $\Delta_{\mathrm{NMI}} = 5.81$ pp; $D_{\mathrm{mean}} = 5.68$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.78$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.94$ pp.						
USPS / compatible	USPS	75.50 ± 1.90	66.26 ± 1.70	79.80 ± 1.60	80.86 ± 1.20	9.24
$n = 9298$; Y0, E13, $\gamma = 1$, G; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -9.24$ pp; $\Delta_{\mathrm{NMI}} = 1.06$ pp; $D_{\mathrm{mean}} = 5.15$; documented source; no pixel identity proof. $\Delta_{\mathrm{SD,ACC}} = -0.20$ pp; $\Delta_{\mathrm{SD,NMI}} = -0.40$ pp.						

Single native-best assignment among 50 fixed-G roundings

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR / compatible	AR_common_mask3306	51.30 ± 1.50	51.19	73.20 ± 1.40	73.34	0.14
$n = 840$; Y0, E8, $\gamma = 1$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -0.11$ pp; $\Delta_{\mathrm{NMI}} = 0.14$ pp; $D_{\mathrm{mean}} = 0.12$; UNVERIFIED.						
JAFFE / compatible	JAFFE	81.20 ± 1.60	95.31	84.30 ± 1.60	94.09	14.11
$n = 213$; Y0, E8, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = 14.11$ pp; $\Delta_{\mathrm{NMI}} = 9.79$ pp; $D_{\mathrm{mean}} = 11.95$; documented source; no pixel identity proof.						
ORL / compatible	ORL_att92	76.00 ± 1.60	75.00	88.60 ± 1.50	88.60	1.00
$n = 400$; Y0, E8, $\gamma = 0.001$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.00$ pp; $\Delta_{\mathrm{NMI}} = 0.00$ pp; $D_{\mathrm{mean}} = 0.50$; documented source; no pixel identity proof.						
UMIST / compatible	UMIST_LZUFE574_stratified_seed022	71.10 ± 1.80	70.92	84.10 ± 1.50	84.36	0.26
$n = 564$; Y0, E8, $\gamma = 0.001$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -0.18$ pp; $\Delta_{\mathrm{NMI}} = 0.26$ pp; $D_{\mathrm{mean}} = 0.22$; UNVERIFIED.						
BinAlpha / adapted	BinAlpha_first30	49.40 ± 1.30	48.33	64.10 ± 1.40	62.97	1.13
$n = 780$; Y0, E8, $\gamma = 1000$, G; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -1.07$ pp; $\Delta_{\mathrm{NMI}} = -1.13$ pp; $D_{\mathrm{mean}} = 1.10$; UNVERIFIED.						
BinAlpha / compatible	BinAlpha_random_seed055	49.40 ± 1.30	49.36	64.10 ± 1.40	64.11	0.04
$n = 780$; Y0, E13, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -0.04$ pp; $\Delta_{\mathrm{NMI}} = 0.01$ pp; $D_{\mathrm{mean}} = 0.03$; UNVERIFIED.						
MSRA50 / adapted	MSRA25_EBSA_16x16	63.20 ± 1.10	56.25	72.20 ± 1.40	64.57	7.63
$n = 1799$; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $\Delta_{\mathrm{ACC}} = -6.95$ pp; $\Delta_{\mathrm{NMI}} = -7.63$ pp; $D_{\mathrm{mean}} = 7.29$; UNVERIFIED.						
YaleB / compatible	YaleB_cropped2414_nearest_Y3	49.80 ± 1.60	44.74	46.80 ± 1.30	52.28	5.48

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
$n = 2414$; Y3, E13, $\gamma = 1000$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -5.06$ pp; $\Delta_{\mathrm{NMI}} = 5.48$ pp; $D_{\mathrm{mean}} = 5.27$; documented source; no pixel identity proof.						
USPS / compatible	USPS	75.50 ± 1.90	65.63	79.80 ± 1.60	81.17	9.87
$n = 9298$; Y0, E8, $\gamma = 1$, V; MAX_ITER_REACHED; $\Delta_{\mathrm{ACC}} = -9.87$ pp; $\Delta_{\mathrm{NMI}} = 1.37$ pp; $D_{\mathrm{mean}} = 5.62$; documented source; no pixel identity proof.						

B.5 Separate certified-search availability

The closest subset matches above belong to the locked finite historical-style IRLS track. A complete 50-start execution does not establish optimality. No comparable certified subset-search track was executed, so no certified closest winner is selected. Separate numerical references, including a continuation on the current USPS input, cannot transfer a certificate to a different saved G. Counts below describe the historical-style coverage only; native rows summarize the same 50-start configuration pool but select one assignment, with no across-run SD.

Table 7: Explicit availability by intended eligible pool and endpoint. Every row has machine status UNAVAILABLE_NO_COMPARABLE_CERTIFIED_SEARCH_COVERAGE in certified_search_status.csv; no winner is fabricated. Aliases are shown in the complete count and removed in the canonical count.

Data	Pool	Endpoint	Complete / canonical configs	Unique inputs	Certified closest
AR	Compatible	Mean	41640 / 41520	1384	Unavailable
AR	Compatible	Native	41640 / 41520	1384	Unavailable
BinAlpha	Adapted	Mean	60 / 60	2	Unavailable
BinAlpha	Adapted	Native	60 / 60	2	Unavailable
BinAlpha	Compatible	Mean	3000 / 3000	100	Unavailable
BinAlpha	Compatible	Native	3000 / 3000	100	Unavailable
JAFFE	Compatible	Mean	30 / 30	1	Unavailable
JAFFE	Compatible	Native	30 / 30	1	Unavailable
MSRA50	Adapted	Mean	30 / 30	1	Unavailable
MSRA50	Adapted	Native	30 / 30	1	Unavailable
ORL	Compatible	Mean	30 / 30	1	Unavailable
ORL	Compatible	Native	30 / 30	1	Unavailable
UMIST	Compatible	Mean	31980 / 28980	966	Unavailable
UMIST	Compatible	Native	31980 / 28980	966	Unavailable
USPS	Compatible	Mean	18 / 18	1	Unavailable
USPS	Compatible	Native	18 / 18	1	Unavailable
YaleB	Compatible	Mean	480 / 480	16	Unavailable
YaleB	Compatible	Native	480 / 480	16	Unavailable

B.6 Additional rounding on locked winners

Data	Endpoint	Select ACC	Select NMI	Confirm ACC	Confirm NMI	Status
AR	mean	51.21	72.94	51.14 ± 1.91	72.88 ± 1.03	COMPLETED
BinAlpha	mean	50.13	61.94	50.14 ± 1.37	62.03 ± 1.02	COMPLETED
BinAlpha	mean	49.54	63.46	49.26 ± 2.81	63.36 ± 1.14	COMPLETED
JAFFE	mean	74.75	85.60	73.96 ± 11.26	84.88 ± 5.91	COMPLETED
MSRA50	mean	49.25	61.50	49.67 ± 4.06	61.38 ± 3.82	COMPLETED
ORL	mean	74.63	89.71	74.51 ± 0.69	89.64 ± 0.26	COMPLETED
UMIST	mean	69.97	85.17	70.60 ± 2.13	85.51 ± 0.85	COMPLETED
USPS	mean	66.26	80.86	65.69 ± 2.77	79.73 ± 2.37	COMPLETED
YaleB	mean	44.24	52.61	44.37 ± 0.83	52.72 ± 0.38	COMPLETED
AR	native	51.19	73.34	51.67	77.42	COMPLETED
BinAlpha	native	48.33	62.97	49.62	63.10	COMPLETED
BinAlpha	native	49.36	64.11	50.77	64.58	COMPLETED
JAFFE	native	95.31	94.09	95.31	94.09	COMPLETED
MSRA50	native	56.25	64.57	46.08	62.82	COMPLETED
ORL	native	75.00	88.60	76.50	89.86	COMPLETED

Data	Endpoint	Select ACC	Select NMI	Confirm ACC	Confirm NMI	Status
UMIST	native	70.92	84.36	60.82	79.17	COMPLETED
USPS	native	65.63	81.17	65.63	81.17	COMPLETED
YaleB	native	44.74	52.28	43.37	51.63	COMPLETED

Additional seeds check rounding on the same samples and saved G. Winners are not reselected. These intervals/SD do not establish historical identity, unseen-data generalization or label-free model selection.

B.7 Baseline comparisons

Data	Method	Setting	ACC	NMI	Status
JAFFE	Ridge LDA (adapted)	limit 300	83.85 ± 2.50	87.77 ± 1.53	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	Ridge LDA–SC (adapted)	adapted_sc	59.38 ± 9.77	73.72 ± 5.60	COMPLETED
JAFFE	PCA_{p_0}–SC	pca_p0_sc	69.00 ± 11.08	83.02 ± 5.86	COMPLETED
JAFFE	PCA_r–SC	pca_r_sc	58.64 ± 7.81	73.20 ± 5.04	COMPLETED
JAFFE	Raw SC	raw_sc	68.53 ± 11.15	83.02 ± 5.72	COMPLETED
MSRA25_EBSA_16x16	Ridge LDA (adapted)	limit 300	57.25 ± 4.00	72.24 ± 2.52	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	Ridge LDA–SC (adapted)	adapted_sc	43.61 ± 4.77	54.73 ± 3.91	COMPLETED
MSRA25_EBSA_16x16	PCA_{p_0}–SC	pca_p0_sc	49.54 ± 5.74	62.89 ± 4.11	COMPLETED
MSRA25_EBSA_16x16	PCA_r–SC	pca_r_sc	37.81 ± 3.64	49.97 ± 2.60	COMPLETED
MSRA25_EBSA_16x16	Raw SC	raw_sc	49.14 ± 6.24	62.84 ± 4.72	COMPLETED
ORL_att92	Ridge LDA (adapted)	limit 300	75.37 ± 3.38	88.94 ± 1.53	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	Ridge LDA–SC (adapted)	adapted_sc	49.19 ± 3.76	75.66 ± 2.65	COMPLETED
ORL_att92	PCA_{p_0}–SC	pca_p0_sc	55.44 ± 4.28	80.12 ± 2.25	COMPLETED
ORL_att92	PCA_r–SC	pca_r_sc	55.29 ± 4.36	80.46 ± 2.21	COMPLETED
ORL_att92	Raw SC	raw_sc	56.18 ± 4.19	80.47 ± 2.29	COMPLETED
USPS	Ridge LDA (adapted)	limit 300	70.17 ± 0.11	68.70 ± 0.09	COMPLETED
Stops: PARTITION STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	Ridge LDA–SC (adapted)	adapted_sc	57.79 ± 6.19	61.10 ± 2.66	COMPLETED
USPS	PCA_{p_0}–SC	pca_p0_sc	62.78 ± 7.95	75.52 ± 4.48	COMPLETED
USPS	PCA_r–SC	pca_r_sc	55.93 ± 4.32	65.98 ± 2.83	COMPLETED
USPS	Raw SC	raw_sc	61.74 ± 8.09	74.94 ± 4.49	COMPLETED
YaleB_cropped2414	Ridge LDA (adapted)	limit 300	42.12 ± 1.62	51.32 ± 1.22	COMPLETED
Stops: MAX ITER (32), PARTITION STABLE (18); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	Ridge LDA–SC (adapted)	adapted_sc	35.85 ± 2.62	50.58 ± 1.59	COMPLETED
YaleB_cropped2414	PCA_{p_0}–SC	pca_p0_sc	17.74 ± 1.15	26.37 ± 0.64	COMPLETED
YaleB_cropped2414	PCA_r–SC	pca_r_sc	15.71 ± 1.06	22.62 ± 0.87	COMPLETED
YaleB_cropped2414	Raw SC	raw_sc	18.34 ± 1.28	26.90 ± 0.67	COMPLETED
JAFFE	LDA existing	limit 100	64.44 ± 9.19	78.60 ± 5.08	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	LDA existing	limit 30	64.44 ± 9.19	78.60 ± 5.08	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	LDA existing	limit 300	64.44 ± 9.19	78.60 ± 5.08	COMPLETED

Data	Method	Setting	ACC	NMI	Status
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 100	46.88 ± 5.32	60.43 ± 4.28	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 30	46.87 ± 5.33	60.41 ± 4.30	COMPLETED
Stops: LABELS STABLE (49), MAX ITER (1); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LDA existing	limit 300	46.88 ± 5.32	60.43 ± 4.28	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 100	56.31 ± 4.14	81.00 ± 1.65	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 30	56.31 ± 4.14	81.00 ± 1.65	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	LDA existing	limit 300	56.31 ± 4.14	81.00 ± 1.65	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 100	58.29 ± 6.61	59.93 ± 4.52	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 30	58.61 ± 6.33	60.29 ± 4.21	COMPLETED
Stops: LABELS STABLE (12), MAX ITER (38); failed updates: 0; zero-success runs: 0.					
USPS	LDA existing	limit 300	58.29 ± 6.61	59.93 ± 4.52	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 100	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 300	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	LDA existing	limit 30	9.67 ± 0.82	13.42 ± 1.06	COMPLETED
Stops: LABELS STABLE (50); failed updates: 0; zero-success runs: 0.					
JAFFE	ClustPy reference	limit 300	82.95 ± 1.07	87.83 ± 0.83	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 625; zero-success runs: 50.					
MSRA25_EBSA_16x16	ClustPy reference	limit 300	54.19 ± 3.44	65.42 ± 2.47	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
ORL_att92	ClustPy reference	limit 300	74.22 ± 2.58	87.76 ± 1.11	COMPLETED
Stops: MAX ITER (50); failed updates: 15000; zero-success runs: 50.					
USPS	ClustPy reference	limit 300	67.23 ± 0.14	65.77 ± 0.13	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
YaleB_cropped2414	ClustPy reference	limit 300	10.57 ± 0.75	14.84 ± 1.24	COMPLETED
Stops: SOURCE NMI STABLE (50); failed updates: 0; zero-success runs: 0.					
MSRA25_EBSA_16x16	LLC	0.0625	51.53 ± 0.40	66.90 ± 0.43	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	55.50 ± 1.50	66.52 ± 0.92	COMPLETED
MSRA25_EBSA_16x16	LLC	1.0	58.91 ± 2.11	68.28 ± 1.07	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	58.72 ± 2.74	69.12 ± 1.55	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	59.16 ± 2.37	69.04 ± 1.00	COMPLETED
MSRA25_EBSA_16x16	LLC	0.0625	56.01 ± 2.67	66.20 ± 1.22	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	51.80 ± 3.26	58.84 ± 1.56	COMPLETED
MSRA25_EBSA_16x16	LLC	1.0	53.73 ± 2.57	63.53 ± 1.83	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	52.15 ± 3.09	62.69 ± 1.98	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	52.52 ± 2.73	62.92 ± 1.96	COMPLETED
MSRA25_EBSA_16x16	LLC	0.0625	56.42 ± 2.20	64.75 ± 1.23	COMPLETED
MSRA25_EBSA_16x16	LLC	0.25	52.82 ± 2.55	62.63 ± 1.58	COMPLETED

Data	Method	Setting	ACC	NMI	Status
MSRA25_EBSA_16x16	LLC	1.0	52.88 ± 2.63	63.12 ± 1.85	COMPLETED
MSRA25_EBSA_16x16	LLC	16.0	51.86 ± 3.05	62.73 ± 1.55	COMPLETED
MSRA25_EBSA_16x16	LLC	4.0	53.26 ± 2.84	63.28 ± 1.72	COMPLETED
MSRA25_EBSA_16x16	K-means	kmeans	51.23 ± 4.85	59.81 ± 3.41	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d10	49.68 ± 4.36	56.62 ± 2.86	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d100	51.22 ± 4.92	59.77 ± 3.50	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d110	51.16 ± 4.91	59.70 ± 3.49	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d120	51.17 ± 4.90	59.71 ± 3.47	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d130	51.07 ± 4.83	59.69 ± 3.40	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d140	51.04 ± 4.87	59.62 ± 3.46	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d150	51.08 ± 4.84	59.65 ± 3.39	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d20	50.87 ± 5.31	58.73 ± 3.75	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d30	51.19 ± 4.93	59.57 ± 3.62	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d40	51.43 ± 4.51	59.68 ± 3.37	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d50	51.27 ± 4.73	59.69 ± 3.45	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d60	51.12 ± 4.74	59.50 ± 3.44	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d70	51.28 ± 4.89	59.74 ± 3.41	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d80	51.21 ± 4.91	59.76 ± 3.53	COMPLETED
MSRA25_EBSA_16x16	PCA–K-means	pca_kmeans__d90	51.22 ± 4.87	59.70 ± 3.51	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d10	34.55 ± 2.85	45.37 ± 2.60	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d100	49.54 ± 5.74	62.89 ± 4.11	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d110	48.93 ± 5.94	63.99 ± 5.10	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d120	48.42 ± 4.57	62.33 ± 3.33	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d130	45.67 ± 4.62	60.11 ± 3.56	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d140	46.56 ± 5.48	59.72 ± 4.17	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d150	45.73 ± 4.63	58.83 ± 3.40	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d20	43.63 ± 4.56	57.19 ± 3.12	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d30	46.00 ± 4.68	59.01 ± 3.38	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d40	45.56 ± 3.88	59.90 ± 2.84	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d50	48.51 ± 5.07	62.75 ± 3.42	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d60	46.86 ± 4.02	61.25 ± 2.84	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d70	47.86 ± 5.57	60.95 ± 5.14	COMPLETED
MSRA25_EBSA_16x16	PCA–SC	pca_sc__d80	49.23 ± 4.15	62.78 ± 3.67	COMPLETED

Data	Method	Setting	ACC	NMI	Status
MSRA25_ EBSA_16x16	PCA–SC	pca_sc__d90	44.87 ± 4.91	59.15 ± 3.93	COMPLETED
MSRA25_ EBSA_16x16	K-means	kmeans	51.23 ± 4.85	59.81 ± 3.41	COMPLETED
JAFFE	SLPS	k5__alpha0.2__ d110	72.21 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.2__ d30	72.20 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.2__ d70	72.21 ± 8.46	84.78 ± 4.19	COMPLETED
JAFFE	SLPS	k5__alpha0.5__ d110	74.41 ± 9.72	85.63 ± 4.45	COMPLETED
JAFFE	SLPS	k5__alpha0.5__ d30	74.45 ± 9.90	85.52 ± 4.58	COMPLETED
JAFFE	SLPS	k5__alpha0.5__ d70	74.41 ± 9.72	85.63 ± 4.45	COMPLETED
JAFFE	SLPS	k5__alpha0.8__ d110	73.06 ± 7.18	84.07 ± 4.05	COMPLETED
JAFFE	SLPS	k5__alpha0.8__ d30	73.13 ± 7.24	84.18 ± 4.02	COMPLETED
JAFFE	SLPS	k5__alpha0.8__ d70	72.85 ± 6.89	83.84 ± 3.96	COMPLETED
JAFFE	SLPS	k5__alpha0.95_ _d70__ SKLEARN_ GREEDY_ KMEANSPP	81.53 ± 6.24	86.97 ± 3.32	COMPLETED
JAFFE	SLPS	k7__alpha0.2__ d110	73.33 ± 8.76	85.46 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.2__ d30	73.11 ± 8.97	85.36 ± 4.24	COMPLETED
JAFFE	SLPS	k7__alpha0.2__ d70	73.33 ± 8.76	85.46 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.5__ d110	74.21 ± 8.75	85.71 ± 4.05	COMPLETED
JAFFE	SLPS	k7__alpha0.5__ d30	74.01 ± 8.37	85.59 ± 3.83	COMPLETED
JAFFE	SLPS	k7__alpha0.5__ d70	74.21 ± 8.75	85.71 ± 4.05	COMPLETED
JAFFE	SLPS	k7__alpha0.8__ d110	73.92 ± 8.02	84.72 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.8__ d30	74.58 ± 8.28	85.14 ± 4.32	COMPLETED
JAFFE	SLPS	k7__alpha0.8__ d70	73.92 ± 8.02	84.72 ± 4.15	COMPLETED
JAFFE	SLPS	k7__alpha0.95_ _d70__ SKLEARN_ GREEDY_ KMEANSPP	80.85 ± 6.58	86.91 ± 3.58	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.2__ d110	52.30 ± 5.18	67.22 ± 3.91	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.2__ d30	52.62 ± 5.08	67.45 ± 3.85	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.2__ d70	52.30 ± 5.17	67.19 ± 3.88	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.5__ d110	52.10 ± 5.41	64.55 ± 4.36	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.5__ d30	52.17 ± 5.23	64.46 ± 4.37	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.5__ d70	52.16 ± 5.31	64.54 ± 4.27	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.8__ d110	51.42 ± 5.00	61.63 ± 3.81	COMPLETED
MSRA25_ EBSA_16x16	SLPS	k5__alpha0.8__ d30	51.18 ± 5.05	61.33 ± 3.83	COMPLETED

Data	Method	Setting	ACC	NMI	Status
MSRA25_EBSA_16x16	SLPS	k5__alpha0.8__d70	51.43 ± 4.96	61.70 ± 3.77	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d110	52.39 ± 5.28	67.77 ± 3.87	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d30	52.48 ± 4.91	67.72 ± 3.72	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.2__d70	52.30 ± 5.15	67.69 ± 3.76	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d110	52.10 ± 5.27	65.53 ± 4.35	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d30	52.37 ± 5.40	65.75 ± 4.49	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.5__d70	52.08 ± 5.32	65.60 ± 4.36	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d110	51.66 ± 5.06	62.58 ± 3.67	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d30	51.71 ± 5.30	62.75 ± 3.88	COMPLETED
MSRA25_EBSA_16x16	SLPS	k7__alpha0.8__d70	51.87 ± 5.05	62.79 ± 3.79	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d110	66.22 ± 3.56	87.15 ± 0.92	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d30	66.65 ± 3.44	87.26 ± 0.95	COMPLETED
ORL_att92	SLPS	k5__alpha0.2__d70	66.29 ± 3.62	87.21 ± 0.92	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d110	65.90 ± 3.91	86.84 ± 1.21	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d30	66.13 ± 3.77	86.91 ± 1.17	COMPLETED
ORL_att92	SLPS	k5__alpha0.5__d70	66.07 ± 3.88	86.92 ± 1.19	COMPLETED
ORL_att92	SLPS	k5__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP	74.68 ± 3.12	88.45 ± 1.19	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d110	64.91 ± 3.18	85.63 ± 1.06	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d30	66.01 ± 3.32	86.12 ± 1.05	COMPLETED
ORL_att92	SLPS	k5__alpha0.8__d70	65.20 ± 3.15	85.78 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d110	64.53 ± 3.72	85.88 ± 1.09	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d30	64.44 ± 3.77	85.78 ± 1.08	COMPLETED
ORL_att92	SLPS	k7__alpha0.2__d70	64.45 ± 3.77	85.90 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d110	65.19 ± 3.72	85.82 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d30	65.51 ± 3.77	85.81 ± 1.11	COMPLETED
ORL_att92	SLPS	k7__alpha0.5__d70	65.26 ± 3.76	85.82 ± 1.08	COMPLETED
ORL_att92	SLPS	k7__alpha0.85__d40__SKLEARN_GREEDY_KMEANSPP	74.19 ± 1.98	87.90 ± 0.87	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d110	64.61 ± 3.01	85.38 ± 1.10	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d30	65.57 ± 3.40	85.58 ± 1.14	COMPLETED
ORL_att92	SLPS	k7__alpha0.8__d70	64.59 ± 2.78	85.40 ± 1.06	COMPLETED

Data	Method	Setting	ACC	NMI	Status
YaleB_cropped2414	SLPS	k5__alpha0.2__d110	11.83 ± 1.07	18.36 ± 1.01	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d30	11.76 ± 1.02	18.28 ± 0.97	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d70	11.83 ± 1.06	18.37 ± 0.96	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	12.11 ± 0.66	18.46 ± 0.62	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.5__d110	10.98 ± 0.92	16.51 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.5__d30	10.83 ± 0.77	16.29 ± 0.75	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.5__d70	10.96 ± 0.87	16.45 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d110	9.91 ± 0.75	14.04 ± 0.85	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d30	9.74 ± 0.65	13.90 ± 0.82	COMPLETED
YaleB_cropped2414	SLPS	k5__alpha0.8__d70	9.88 ± 0.70	14.01 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d110	11.13 ± 1.01	16.95 ± 0.94	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d30	11.11 ± 1.01	16.92 ± 0.93	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d70	11.13 ± 1.03	16.94 ± 0.93	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	11.24 ± 0.54	17.07 ± 0.44	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d110	10.79 ± 0.89	16.02 ± 0.89	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d30	10.70 ± 0.89	15.96 ± 0.87	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.5__d70	10.82 ± 0.86	16.05 ± 0.86	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d110	10.09 ± 0.78	14.22 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d30	9.94 ± 0.73	14.08 ± 0.84	COMPLETED
YaleB_cropped2414	SLPS	k7__alpha0.8__d70	10.10 ± 0.77	14.21 ± 0.81	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d110	13.07 ± 1.17	20.48 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d30	12.90 ± 1.10	20.32 ± 1.06	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d70	13.05 ± 1.19	20.54 ± 1.13	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	14.42 ± 0.73	21.36 ± 0.72	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d110	11.81 ± 0.95	18.17 ± 0.91	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d30	11.51 ± 0.93	17.76 ± 0.82	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.5__d70	11.76 ± 0.93	18.07 ± 0.88	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.8__d110	10.51 ± 0.80	15.06 ± 0.79	COMPLETED
YaleB_dengcai32	SLPS	k5__alpha0.8__d30	10.18 ± 0.80	14.70 ± 0.88	COMPLETED

Data	Method	Setting	ACC	NMI	Status
YaleB_dengcai32	SLPS	k5__alpha0.8__d70	10.40 ± 0.82	14.90 ± 0.73	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d110	12.36 ± 1.23	18.90 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d30	12.24 ± 1.20	18.71 ± 1.11	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d70	12.33 ± 1.23	18.83 ± 1.08	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.2__d70__SKLEARN_GREEDY_KMEANSPP	13.52 ± 0.78	19.44 ± 0.69	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.5__d110	11.85 ± 1.02	17.74 ± 0.90	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.5__d30	11.61 ± 0.99	17.51 ± 0.89	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.5__d70	11.87 ± 0.99	17.79 ± 0.89	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d110	10.65 ± 0.79	15.35 ± 0.75	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d30	10.48 ± 0.79	15.15 ± 0.86	COMPLETED
YaleB_dengcai32	SLPS	k7__alpha0.8__d70	10.66 ± 0.83	15.36 ± 0.74	COMPLETED

B.8 USPS numerical audit on one newly frozen F/W pair

These rows use a newly generated, recoverable current USPS representation and its exact shrinkage graph. They do not identify or replace the missing historical selected F/G arrays. Every non-SC point targets unsquared E8 with undirected coefficient $2\gamma w$. All original ten stages remain visible; the two additional restarts are separate numerical attempts, not replacements for the historical finite-iteration winner or subset-search output.

The numerical audit and subset USPS representation record exactly matching current F/W content addresses. Recorded input-reference status: MATCHING_RECORDED_CURRENT_F_W_OBJECTS. This is only a comparison of current input references. Each certificate belongs to its own saved continuation G; it does not certify the locked IRLS G, establish a certified closest search result, or recover missing historical F/G arrays.

Tables 10 and 11 refer to the same stage/gamma rows. C/U denote the recorded certified/uncertified state, with a required relative gap at most 10^{-5} and capacity violation at most 10^{-8} for certification. A solver stopping status alone is not a certificate. Uncertified points cannot establish that the mathematical objective lacks clustering capacity. Missing exports or saved points stay unavailable.

Table 10: Same-target numerical evidence. Original rows show fit CPU only where recorded; extra-restart CPU includes the full numerical attempt. A dash means unavailable metadata, not zero cost.

Stage	γ	Primal	Dual	Rel. gap	Cert.	CPU (s)	Solver status
SC on F	–	–	–	–	–	–	NOT_APPLICABLE
IRLS300	1	13.577	–	–	U	1800.0	CPU_LIMIT
Uniform	1	13.644	13.562	0.006	U	160.0	MAX_ITER_REACHED
Diagonal	1	13.587	13.576	0.000864	U	160.5	MAX_ITER_REACHED
Flow	1	267.96	1.6424e-16	1	U	12.7	INCOMPLETE_LINEAR_SOLVE
Warm restart	1	13.577	13.577	2.58e-05	U	155.2	MAX_ITER_REACHED
IRLS300	1000	388.48	–	–	U	11.7	FAILED

Stage	γ	Primal	Dual	Rel. gap	Cert.	CPU (s)	Solver status
Uniform	1000	267.96	87.328	0.674	U	154.2	MAX_ITER_ REACHED
Diagonal	1000	267.96	86.357	0.678	U	142.7	MAX_ITER_ REACHED
Flow	1000	267.96	1.6424e-13	1	U	11.3	INCOMPLETE_ LINEAR_SOLVE
Warm restart	1000	267.96	120.26	0.551	U	136.5	MAX_ITER_ REACHED
Extra restart	1	13.577	13.577	1e-05	C	269.4	PRIMAL_DUAL_ GAP_CERTIFIED
Extra restart	1000	267.96	176.38	0.342	U	811.4	MAX_ITER_ REACHED

Clustering metrics below are percent, mean $\pm$ sample SD over the same fifty greedy K-means++/Lloyd seeds 0–49 on each saved G. Both metrics come from one complete endpoint. This is conditional rounding variability, not independent datasets or full-pipeline uncertainty. A native-best assignment is not substituted for a mean, and its single-run value receives no across-run SD.

Table 11: Paired rounding protocol on the fixed current F/W. The IRLS300 flag refers only to completion of the declared historical-style 300-step protocol, including its iterate stopping rule; it does not certify optimality. Other stages are separate algorithms/restarts.

Stage	γ	ACC	NMI	n	IRLS300 complete	Evaluation status
SC on F	–	65.60 ± 1.87	80.31 ± 2.13	50	–	COMPLETED
IRLS300	1	65.56 ± 2.52	80.23 ± 2.34	50	no	COMPLETED
Uniform	1	65.40 ± 2.48	80.11 ± 2.28	50	–	COMPLETED
Diagonal	1	65.57 ± 2.36	80.08 ± 2.28	50	–	COMPLETED
Flow	1	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Warm restart	1	65.56 ± 2.52	80.23 ± 2.34	50	–	COMPLETED
IRLS300	1000	77.37 ± 4.49	84.84 ± 1.56	50	no	COMPLETED
Uniform	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Diagonal	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Flow	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Warm restart	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED
Extra restart	1	65.56 ± 2.52	80.23 ± 2.34	50	–	COMPLETED
Extra restart	1000	16.70 ± 0.00	0.00 ± 0.00	50	–	COMPLETED

B.9 Fixed-representation graph evidence: G1–G3

This section uses the preregistered full ORL and cropped YaleB inputs, independently of retrospective subset winners. No best gamma, budget, graph or seed is selected here. Means and sample SD use the same fifty paired greedy K-means++/Lloyd seeds 0–49. A dash denotes unavailable evidence. C/U records certified/uncertified solver status; certification does not imply accurate clustering. The source files and hashes are listed in the generated report-input manifest.

The graph design contains 362 recorded comparisons across G1, G2 and G3. Scientific outcome and numerical qualification remain distinct.

Table 12: Graph-design coverage, including unsuccessful and unevaluated configurations. Shared mathematical problem identities are aliases, not independent measurements.

Phase	Observation	State	Tasks
G1	Scientific output	CASEBOOK_COMPLETE	2
G2	Scientific output	POINT_COMPLETE	48
G3	Scientific output	POINT_COMPLETE	312

G1: fixed gamma 1 casebooks. Casebook correctness uses an optimal assignment separately for each partition. Error-rate means below average the fifty paired rounding outcomes; deltas are SSC minus SC. They describe the fixed data and representation, not independent test datasets.

Table 13: Matched G1 means and sample SD at fixed gamma 1; these are not selected gamma-path maxima.

Data	SC ACC	SSC ACC	SC NMI	SSC NMI	SSC solver
ORL	78.33 ± 1.31	78.48 ± 1.41	89.87 ± 0.67	89.89 ± 0.80	PRIMAL_DUAL_GAP_ CERTIFIED
YaleB cropped	21.78 ± 0.45	21.51 ± 0.48	27.73 ± 0.58	27.45 ± 0.58	PRIMAL_DUAL_GAP_ CERTIFIED

Table 14: False merges and splits use contingency-count definitions. The averaging unit is a paired rounding seed, not a sample pair treated as an independent replicate.

Data	Pair error	SC (%)	SSC (%)	Delta (pp)	Pairs of rounds
ORL	False merge	0.91	0.92	0.01	50
ORL	False split	24.37	24.21	-0.16	50
YaleB cropped	False merge	7.78	7.95	0.17	50
YaleB cropped	False split	84.31	84.34	0.02	50

Table 15: Upstream A and shrinkage W0 contamination remain distinct. An unavailable A field is not replaced with W0 evidence. Fragmentation is class-induced and includes isolates.

Data	Graph / length space	Edges	Cross edges (%)	Cross mass (%)	Excess fragments
ORL	A: input-feature geometry	1273	22.31	11.99	8
ORL	W0: spectral geometry	1217	17.01	4.65	16
YaleB cropped	A: input-feature geometry	8510	58.88	53.60	438
YaleB cropped	W0: spectral geometry	7957	63.24	70.23	1109

Table 16: Predefined row-norm and weighted-degree quintiles, with original cut points and tie handling. Fractions pool the recorded sample-by-round outcomes within each fixed stratum.

Data	Variable	Quintile	n	Wrong degree	Wrong mass	SC right/SSC wrong (%)	SC wrong/SSC right (%)
ORL	row_norm	0	80	2.60	0.073	8.70	10.53
ORL	row_norm	1	80	1.06	0.118	5.78	6.17
ORL	row_norm	2	80	0.62	0.123	3.77	3.17
ORL	row_norm	3	80	0.36	0.160	4.23	4.03
ORL	row_norm	4	80	0.53	0.041	9.12	8.48
ORL	weighted_ degree	0	80	2.33	0.021	10.25	10.57
ORL	weighted_ degree	1	80	1.52	0.112	7.58	9.62
ORL	weighted_ degree	2	80	0.60	0.116	5.88	5.58
ORL	weighted_ degree	3	80	0.30	0.056	4.15	2.85
ORL	weighted_ degree	4	80	0.42	0.211	3.75	3.75
YaleB cropped	row_norm	0	483	5.48	2.305	1.39	1.67
YaleB cropped	row_norm	1	483	5.19	2.347	3.14	2.79
YaleB cropped	row_norm	2	482	4.91	2.002	5.10	5.59
YaleB cropped	row_norm	3	483	3.83	1.170	7.89	6.15
YaleB cropped	row_norm	4	483	1.43	0.363	2.39	2.36

Data	Variable	Quintile	n	Wrong degree	Wrong mass	SC right/SSC wrong (%)	SC wrong/SSC right (%)
YaleB cropped	weighted_ degree	0	483	2.58	0.157	4.94	4.85
YaleB cropped	weighted_ degree	1	483	3.13	0.618	4.89	4.19
YaleB cropped	weighted_ degree	2	482	4.09	1.255	4.23	4.07
YaleB cropped	weighted_ degree	3	483	4.62	2.070	3.40	3.42
YaleB cropped	weighted_ degree	4	483	6.42	4.087	2.45	2.02

Table 17: Deterministic first original row in each fixed seed-zero correctness category. Missing categories stay unavailable; no private image is embedded.

Data	Seed-zero category	Original row ID	Seed
ORL	sc_correct_ssc_wrong	1	0
ORL	sc_wrong_ssc_correct	271	0
ORL	both_correct	10	0
ORL	both_wrong	0	0
YaleB cropped	sc_correct_ssc_wrong	145	0
YaleB cropped	sc_wrong_ssc_correct	174	0
YaleB cropped	both_correct	0	0
YaleB cropped	both_wrong	2	0

G2: all 48 fixed cells. F0 is the saved raw spectral anchor; U is its row-normalized version. W0 and Wa are constructed in those respective geometries. Every graph is mass-matched to W0; equal mass does not equalize node forces or movement. Gamma zero is the anchor-only SC endpoint. Cached gamma-zero graph-arm rows are declared aliases, not independent fits.

Table 18: Complete fixed G2 path, including unavailable/failed cells. Movement is sum of row displacements divided by sum of anchor row norms; it is not Frobenius movement.

Data	Anchor	Graph	γ	ACC	NMI	Cert.	L21 move	Scientific outcome
ORL	F0	W0	0	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	W0	0.001	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	W0	1	78.48 ± 1.41	89.89 ± 0.80	C	0.0896	POINT_COMPLETE
ORL	F0	W0	10	77.68 ± 1.43	90.44 ± 0.56	C	0.15	POINT_COMPLETE
ORL	F0	W0	1000	74.52 ± 0.25	89.66 ± 0.18	U	0.302	POINT_COMPLETE
ORL	F0	W0	1e+06	38.25 ± 0.00	66.35 ± 0.00	U	0.678	POINT_COMPLETE
ORL	F0	Wa	0	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	Wa	0.001	78.33 ± 1.31	89.87 ± 0.67	C	0	POINT_COMPLETE
ORL	F0	Wa	1	77.83 ± 1.29	89.33 ± 0.69	C	0.0819	POINT_COMPLETE
ORL	F0	Wa	10	77.88 ± 1.21	89.49 ± 0.67	C	0.105	POINT_COMPLETE
ORL	F0	Wa	1000	78.07 ± 1.14	89.58 ± 0.60	C	0.132	POINT_COMPLETE
ORL	F0	Wa	1e+06	77.74 ± 1.32	89.59 ± 0.60	U	0.148	POINT_COMPLETE
ORL	U	W0	0	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	W0	0.001	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	W0	1	80.01 ± 1.54	90.94 ± 0.58	C	0.0452	POINT_COMPLETE
ORL	U	W0	10	79.92 ± 1.49	90.95 ± 0.40	C	0.0882	POINT_COMPLETE
ORL	U	W0	1000	74.82 ± 0.24	89.85 ± 0.09	U	0.257	POINT_COMPLETE
ORL	U	W0	1e+06	72.78 ± 0.76	89.35 ± 0.15	U	0.537	POINT_COMPLETE
ORL	U	Wa	0	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	Wa	0.001	80.17 ± 1.55	90.89 ± 0.64	C	0	POINT_COMPLETE
ORL	U	Wa	1	80.34 ± 1.45	90.97 ± 0.53	C	0.0213	POINT_COMPLETE
ORL	U	Wa	10	80.33 ± 1.44	90.94 ± 0.57	C	0.0374	POINT_COMPLETE
ORL	U	Wa	1000	80.01 ± 1.51	90.89 ± 0.58	C	0.0578	POINT_COMPLETE
ORL	U	Wa	1e+06	80.14 ± 1.42	90.88 ± 0.61	U	0.0718	POINT_COMPLETE
YaleB cropped	F0	W0	0	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE

Data	Anchor	Graph	γ	ACC	NMI	Cert.	L21 move	Scientific outcome
YaleB cropped	F0	W0	0.001	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	W0	1	21.51 ± 0.48	27.45 ± 0.58	C	0.0976	POINT_COMPLETE
YaleB cropped	F0	W0	10	21.60 ± 0.41	27.97 ± 0.33	U	0.335	POINT_COMPLETE
YaleB cropped	F0	W0	1000	16.88 ± 0.17	24.76 ± 0.61	U	0.714	POINT_COMPLETE
YaleB cropped	F0	W0	1e+06	9.82 ± 0.00	21.40 ± 0.00	U	0.838	POINT_COMPLETE
YaleB cropped	F0	Wa	0	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	Wa	0.001	21.78 ± 0.45	27.73 ± 0.58	C	0	POINT_COMPLETE
YaleB cropped	F0	Wa	1	22.53 ± 0.42	28.08 ± 0.53	C	0.16	POINT_COMPLETE
YaleB cropped	F0	Wa	10	23.37 ± 0.45	28.93 ± 0.60	U	0.274	POINT_COMPLETE
YaleB cropped	F0	Wa	1000	3.40 ± 0.00	8.67 ± 0.00	U	0.964	POINT_COMPLETE
YaleB cropped	F0	Wa	1e+06	3.40 ± 0.00	8.67 ± 0.00	U	0.964	POINT_COMPLETE
YaleB cropped	U	W0	0	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	W0	0.001	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	W0	1	25.17 ± 0.85	31.88 ± 0.46	C	0.117	POINT_COMPLETE
YaleB cropped	U	W0	10	24.56 ± 0.50	30.91 ± 0.28	C	0.393	POINT_COMPLETE
YaleB cropped	U	W0	1000	15.08 ± 0.00	30.02 ± 0.00	U	0.82	POINT_COMPLETE
YaleB cropped	U	W0	1e+06	9.82 ± 0.00	21.40 ± 0.00	U	0.897	POINT_COMPLETE
YaleB cropped	U	Wa	0	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	Wa	0.001	25.80 ± 0.71	32.59 ± 0.43	C	0	POINT_COMPLETE
YaleB cropped	U	Wa	1	26.11 ± 0.77	32.68 ± 0.53	C	0.0494	POINT_COMPLETE
YaleB cropped	U	Wa	10	26.49 ± 0.92	32.81 ± 0.62	C	0.165	POINT_COMPLETE
YaleB cropped	U	Wa	1000	28.30 ± 0.50	34.71 ± 0.32	U	0.447	POINT_COMPLETE
YaleB cropped	U	Wa	1e+06	3.40 ± 0.00	8.67 ± 0.00	U	0.957	POINT_COMPLETE

G3: fixed primary budget and gamma. The following rows use only the 15% addition budget and effective E8 gamma 1000. Additions come from the already frozen label-blind pool, with at most two newly incident edges per node and median positive W0 edge weight before mass matching. Oracle additions join same-class components after cross-class deletion and are reused unchanged in add-only and both arms. Other gammas and the 5% budget remain sensitivity records in the full CSVs, not alternative promotion gates.

Table 19: Fixed primary G3 arms. Blind controls retain exactly the Oracle deletion mask and randomize only additions. Same-class random additions do not prefer separate components. Fully randomized deletion/addition is a separate supplementary control. Infeasible or unrun controls retain their status.

Data	Arm / control	ACC	NMI	Cert.	L21 move	Excess fragments	Outcome
ORL	No edits	74.52 ± 0.25	89.66 ± 0.18	U	0.302	16	POINT_ COMPLETE
ORL	Oracle delete	82.92 ± 1.22	92.95 ± 0.37	C	0.183	16	POINT_ COMPLETE
ORL	Oracle add	68.57 ± 0.21	90.40 ± 0.04	U	0.457	1	POINT_ COMPLETE
ORL	Oracle delete+completion	93.34 ± 1.15	97.98 ± 0.33	C	0.302	1	POINT_ COMPLETE
ORL	Fixed delete + blind add 400	60.92 ± 0.22	86.70 ± 0.18	C	0.4	16	POINT_ COMPLETE
ORL	Fixed delete + blind add 401	73.80 ± 0.67	91.57 ± 0.28	C	0.355	15	POINT_ COMPLETE
ORL	Fixed delete + blind add 402	70.71 ± 0.17	89.79 ± 0.18	C	0.39	15	POINT_ COMPLETE
ORL	Fixed delete + same-class random 400	87.19 ± 1.24	95.28 ± 0.37	C	0.244	11	POINT_ COMPLETE
ORL	Fixed delete + same-class random 401	85.97 ± 1.00	94.44 ± 0.24	C	0.226	13	POINT_ COMPLETE
ORL	Fixed delete + same-class random 402	86.24 ± 1.34	94.90 ± 0.31	C	0.228	13	POINT_ COMPLETE
ORL	Fully random delete/add 400	57.62 ± 0.62	83.55 ± 0.35	U	0.553	15	POINT_ COMPLETE
ORL	Fully random delete/add 401	68.11 ± 0.44	87.83 ± 0.19	U	0.516	19	POINT_ COMPLETE
ORL	Fully random delete/add 402	61.64 ± 0.96	84.51 ± 0.40	U	0.568	17	POINT_ COMPLETE
YaleB cropped	No edits	16.88 ± 0.17	24.76 ± 0.61	U	0.714	1109	POINT_ COMPLETE
YaleB cropped	Oracle delete	23.17 ± 0.64	30.63 ± 0.71	U	0.173	1109	POINT_ COMPLETE
YaleB cropped	Oracle add	11.64 ± 0.49	17.43 ± 0.83	C	0.869	53	POINT_ COMPLETE
YaleB cropped	Oracle delete+completion	54.04 ± 1.85	80.29 ± 0.84	U	0.764	53	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 400	6.04 ± 0.07	11.96 ± 0.20	U	0.762	1000	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 401	6.03 ± 0.10	11.63 ± 0.21	U	0.76	1023	POINT_ COMPLETE
YaleB cropped	Fixed delete + blind add 402	5.81 ± 0.10	11.58 ± 0.19	U	0.752	1016	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 400	41.48 ± 1.75	51.96 ± 0.93	C	0.532	661	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 401	42.25 ± 1.71	53.22 ± 0.79	C	0.538	640	POINT_ COMPLETE
YaleB cropped	Fixed delete + same-class random 402	40.96 ± 1.47	51.36 ± 0.71	C	0.517	684	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 400	4.20 ± 0.05	9.70 ± 0.11	U	0.936	1376	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 401	4.20 ± 0.05	10.02 ± 0.09	U	0.936	1368	POINT_ COMPLETE
YaleB cropped	Fully random delete/add 402	4.60 ± 0.06	10.57 ± 0.13	U	0.933	1380	POINT_ COMPLETE

Table 20: Preregistered promotion checks. Certification and activity are required for delete and both; blind-control certificate flags remain visible but are not an additional hidden pass condition. At least one development dataset must pass.

Data	Locked gate condition	Threshold	Satisfied	Decision
ORL	Delete and both certified	gap <= 1e-5; capacities <= 1e-8	yes	PASSED
ORL	Delete and both active	normalized L21 > 1e-6	yes	PASSED
ORL	Both minus delete ACC	mean gain >= 1 pp	yes	PASSED
ORL	Both minus delete NMI	mean gain >= 0 pp	yes	PASSED
ORL	Class-induced excess fragments	reduction >= 20 percent	yes	PASSED
ORL	ACC CI versus delete	lower endpoint > 0 pp	yes	PASSED
ORL	ACC CI versus mean blind additions	lower endpoint > 0 pp	yes	PASSED
ORL	Recorded excess-fragment reduction	93.75 percent	–	PASSED
YaleB cropped	Delete and both certified	gap <= 1e-5; capacities <= 1e-8	no	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	Delete and both active	normalized L21 > 1e-6	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	Both minus delete ACC	mean gain >= 1 pp	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	Both minus delete NMI	mean gain >= 0 pp	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	Class-induced excess fragments	reduction >= 20 percent	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	ACC CI versus delete	lower endpoint > 0 pp	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	ACC CI versus mean blind additions	lower endpoint > 0 pp	yes	NOT_EVALUABLE_UNCERTIFIED
YaleB cropped	Recorded excess-fragment reduction	95.22 percent	–	NOT_EVALUABLE_UNCERTIFIED

Table 21: Stored paired percentile intervals: 10,000 resamples, bootstrap seed 424242. The three random-control outcomes are averaged within each rounding seed before the contrast. Only ACC intervals versus deletion and the mean blind control enter the gate; supplementary controls and NMI intervals are reported without redefining it.

Data	Fixed contrast	Metric	Mean delta (pp)	95% lower	95% upper
ORL	Both minus delete	ACC	10.41	9.92	10.91
ORL	Both minus delete	NMI	5.02	4.87	5.18
ORL	Both minus fixed-delete / blind-add mean	ACC	24.86	24.56	25.17
ORL	Both minus fixed-delete / blind-add mean	NMI	8.62	8.53	8.71
ORL	Both minus fixed-delete / same-class-random mean	ACC	6.87	6.53	7.23
ORL	Both minus fixed-delete / same-class-random mean	NMI	3.10	3.00	3.21
ORL	Both minus fully random edits mean	ACC	30.88	30.58	31.19
ORL	Both minus fully random edits mean	NMI	12.68	12.57	12.79
YaleB cropped	Both minus delete	ACC	30.87	30.39	31.34
YaleB cropped	Both minus delete	NMI	49.66	49.34	49.97
YaleB cropped	Both minus fixed-delete / blind-add mean	ACC	48.07	47.57	48.57

Data	Fixed contrast	Metric	Mean delta (pp)	95% lower	95% upper
YaleB cropped	Both minus fixed-delete / blind-add mean	NMI	68.57	68.34	68.81
YaleB cropped	Both minus fixed-delete / same-class-random mean	ACC	12.47	11.92	13.04
YaleB cropped	Both minus fixed-delete / same-class-random mean	NMI	28.11	27.87	28.36
YaleB cropped	Both minus fully random edits mean	ACC	49.70	49.19	50.20
YaleB cropped	Both minus fully random edits mean	NMI	70.20	69.96	70.44

B.10 Coverage, solver qualification and cost

Complete-config input coverage means at least one completed SSC configuration on an input, not a finished search design. 2482 unique inputs yielded 74436 complete SSC input/configuration pairs; 2480 inputs completed all 30 SSC points and 2480 completed all 30 SSC plus three matched-SC points.

Data	Intended aliases	Unique inputs	Measured inputs	Complete configurations
AR	3537	3533	1385	41670
JAFFE	1	1	1	30
ORL	1	1	1	30
UMIST	1073	971	971	32190
BinAlpha	104	104	104	3120
MSRA50	1	1	1	30
YaleB	16	16	16	480
USPS	3	3	3	66

Execution status	Intended combinations
COMPLETED	74436
COMPLETED_ALIAS	3180
FAILED_COMPLETION_PRESERVED	18
FAILED_ORIGINAL_PRESERVED	6
NOT_RUN	64440

The recorded charge for retained historical study phases and shared overhead is 29.0304 CPU-core hours, including conservative accounting caps for metadata operations. Experiments used at most 2 single-thread workers and an aggregate RSS target of 8.0 GiB. Per-fit measured costs and numerical qualifications are reported separately from clustering performance.

Experiment group	CPU seconds
G1	50.6
G2	1028.5
G3	3535.9
aggregation	3786.8
confirmation	11.7
mandatory_baseline	6852.2
mandatory_repair	2269.9
mandatory_usps	3840.2
mandatory_usps_evaluation	11.3
profile	838.8
subset	81380.5
validation	902.9

Graph criterion. Oracle intervention criterion: PASSED. This is a label-assisted diagnostic on fixed development inputs.

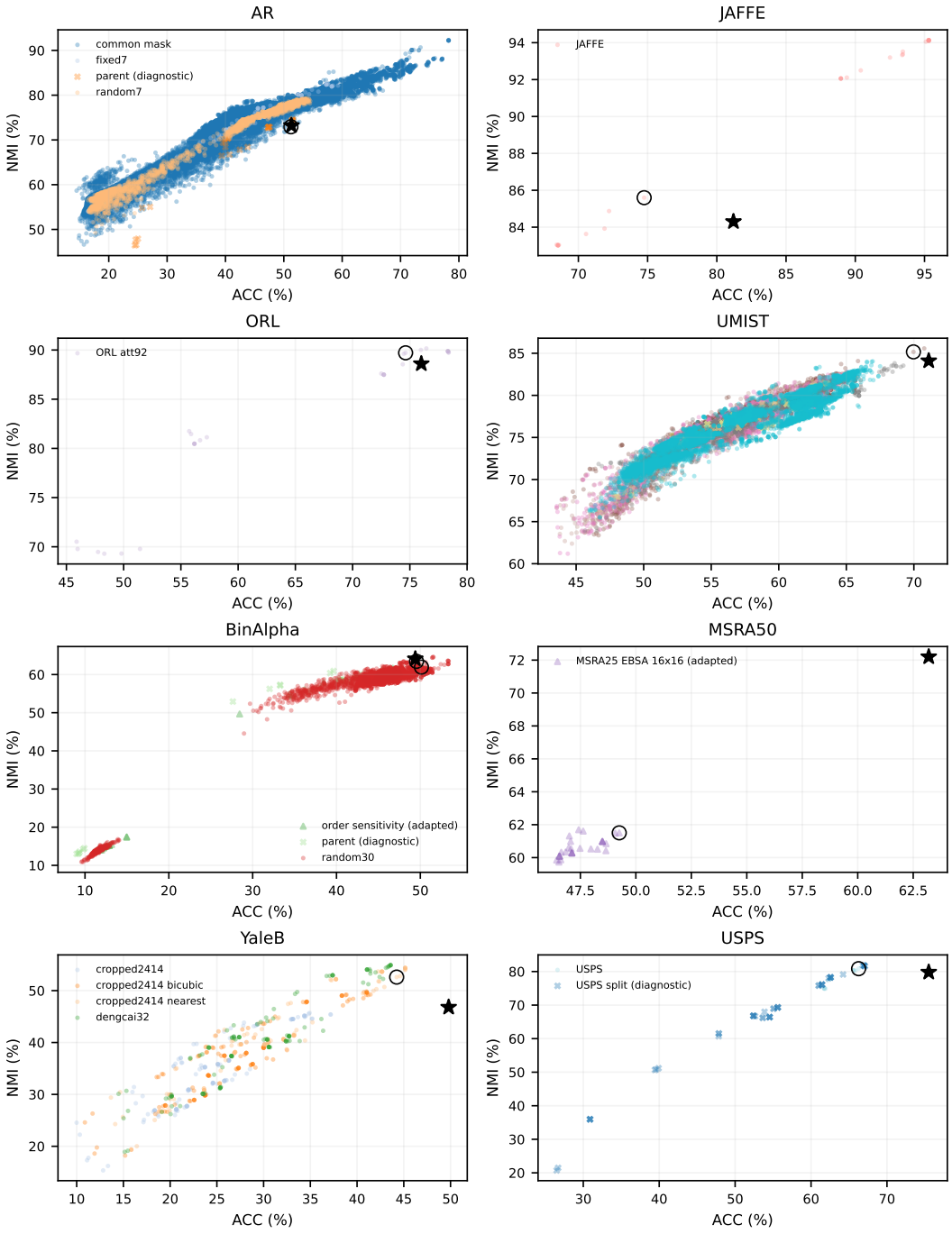


Figure 2: SSC search distribution for the mean/sample-SD endpoint. Points are complete candidate/configuration records; colors distinguish source/subset families (YaleB colors separate source/resize versions), triangles are adapted rows, crosses are diagnostic rows, stars are published SSC targets and outlines mark the already locked selections within each eligible pool. Eligibility pools and actual locked selections are reported in the tables. More candidates can improve a retrospective match without identifying historical data.

Generated-input provenance. Tables and figures are generated from the authoritative committed CSV/JSON records only. The input hash manifest is `results/submission_refresh/report_input_manifest.json`. Generated snapshot date: 2026-11-11.

C Additional protocol and implementation detail

The source/subset catalogue fixes AR seven-of-fourteen rules, UMIST deduplication/omission hypotheses and BinAlpha 30-per-letter draws with explicit original indices. Diagnostic parents cannot compete as equivalent paper reconstructions. ORL/JAFFE membership is not arbitrarily searched. Yale source, resize interpolation and Y0–Y3 normalization factors remain separate; USPS uses all 9,298 rows in compatible experiments. Native 16-by-16 MSRA is labelled as an adapted-resolution source and is not upsampled to claim recovery of the paper's 32-by-32 data. A changed subset always rebuilds both graphs and both representations. Complete coverage, locked selections and confirmation records, rather than numerical closeness alone, determine the reported scope.

The instrumented ClustPy port pins commit `4ac9d00685efc7432800684f5bfac0f01af59172`[1]. It preserves failed-update continuation and both the actually used and source-returned projections. Scoped NumPy seeding of the source PCA is the disclosed determinism modification. The separately named PCA-preconditioned ridge LDA-Kmeans→SC adaptation uses centered float64 input, numerical rank tolerance $\epsilon \max(n,p)\sigma_{\max}$, $p_0 = \min(100, p, n-c, \operatorname{rank}(X_c))$, $r = \min(c-1, p_0)$ and ridge $\eta = 10^{-6}\operatorname{tr}(S_t)/p_0$. Covariance scatter uses divisor $n-1$; inner K-means has ten starts, the outer cap is 300, and stopping compares partitions up to label permutation. The actually clustered projection is returned. Raw SC, PCA_{p_0} SC and PCA_r SC controls use one RANDOM Lloyd start per matched external seed; direct learned-label versus downstream SC contrasts therefore also reflect their prescribed rounding budgets. Neither this adaptation nor a degenerate reference execution recovers historical LDA-SC.

The diagonal PDHG formulation uses $K = \operatorname{diag}(\sqrt{w})B$ and maps its dual back to original capacities $2\gamma w$; independent small convex references validate the target. Agreement between numerical solvers alone is not a certificate. Orthogonal eigenspace rotations preserve Euclidean distances, while a changed cutoff eigenspace or row normalization can change clustering. Edge deletion need not preserve $G = F$: on an eight-node cycle at $\gamma = 0.6$, removing one edge destroys force cancellation. The separately valid incident-mass sufficient bound is $2\gamma \max_i \sum_j w_{ij} \le 1$.

G3 holds the full development F_0 fixed and uses a label-blind top-20-neighbor addition pool for oracle evaluation. The fixed 15%/gamma-1000 gate requires certified active deletion and deletion-plus-completion solves, at least one percentage point paired ACC gain, nondecreasing NMI, at least 20% fewer excess class fragments, and positive conditional bootstrap intervals against deletion and fixed oracle deletion plus mean blind random additions. Fully randomized deletion/addition and same-class random additions remain separate controls. The labelled intervention diagnoses available graph information; it is not a label-free algorithm or an unseen-data test.

D Nonconvex fusion: scope and negative evidence

Truncated fusion is established in PRclust [Pan et al., 2013, Wu et al., 2016]; distance-dependent robust pairwise fusion is central to Shah–Koltun RCC [Shah and Koltun, 2017]. These differ from Sun et al.'s resistant convex clustering [Sun et al., 2025]. The SSC adaptation does not claim invention of these penalties.

For two concentrated groups of sizes $m < n$, separation d and aggregate attraction λ, the unsquared-fidelity linear-fusion optimum is $d\min(m, \lambda)$. With capped penalty $\min(r, \tau)$ and $d > \tau$, it is $\min(md, \lambda\tau)$. Thus $m < \lambda < md/\tau$ gives fusion for the linear objective and separation for the capped objective. Equality can admit multiple minimizers. These two-representative statements are not arbitrary-graph clustering guarantees.

A separately enumerated five-point graph has linear optimum 8 and capped optimum 3. The local capped algorithm obtains 3 from F but 8 from the original SSC output: accurate inner convex solves

[1] `https://github.com/collinleiber/ClustPy/blob/4ac9d00685efc7432800684f5bfac0f01af59172/clustpy/centroid/ldakmeans.py`; the original source, BSD-3-Clause notice and exact hashes are retained.

do not certify a global nonconvex solution. Synthetic imbalanced-bridge experiments also show the limitation of damage avoidance: SSC averages 99.156 ACC/96.725 NMI while capped fusion, static screening and squared fidelity recover 100/100, which SC already achieves. Capping is therefore neither necessary in that example nor evidence of additional clustering capacity over SC. Historical synthetic and secondary geometry results remain distinct from the independent refinement studies.

The mathematical and empirical observations motivate a narrow conclusion: representation motion, robustness or objective improvement alone is insufficient. A useful extension must retain both ACC and NMI improvements over appropriate SC, SSC and mechanism controls, with numerical qualifications attached to the exact saved outputs.

E Independent refinement objectives, bounds and certificates

Let $c_e = 2\gamma w_e$, $a_e = \eta\|(BH)_e\|$ and $J_0(G) = \sum_i \|g_i - h_i\| + \sum_e c_e\|(BG)_e\|$. Every positive undirected edge is counted once. The registered input memberships, preparation, exact arrays, source identities and triplet order are fixed before new scoring. ORL and cropped YaleB are development datasets, not unseen validation samples.

E1: relative stopping distance.

$$J_\eta(G) = \sum_i \|g_i - h_i\| + \sum_e c_e(\|(BG)_e\| - a_e)_+, \qquad D_\eta(q) = \langle H, B^T q\rangle - \sum_e a_e\|q_e\|.$$

The dual constraints are $\|q_e\| \le c_e$ and $\|(B^T q)_i\| \le 1$. With original-edge step σ_e, the dual prox soft-thresholds the norm by $\sigma_e a_e$ then projects to c_e. In weighted-incidence coordinates $K = \mathrm{diag}(\sqrt{w})B$, the mapped original-edge step is $w_e\Sigma_e$. For a geometric median z of H, the candidate $\eta H + (1-\eta)z$ has zero edge penalty and fidelity $(1-\eta)D(H)$. Hence $P^* \le (1-\eta)D(H)$. The triangle inequality gives $D(H) \le \sum_i \|g_i - h_i\| + D(G)$, proving $D(G) \ge \eta D(H) - \epsilon$ for an output with absolute certified gap $\epsilon = P(G) - D_\eta(q)$. Certified median lower/upper bounds determine the numerical margins. These statements concern G, not normalized Q.

A supplementary objective comparison explains when E1 cannot change the ordinary optimum. Let $C = \sum_e c_e a_e$. Since $(r-a)_+ \ge r - a$, $J_\eta(G) \ge J_0(G) - C$. If an exact ordinary minimizer has all edge distances at least a_e, equality holds there and that same representation minimizes E1. Saved-output tube and cross-objective diagnostics inspect this condition without another fit or parameter selection. Finite certificates remain attached to their own targets.

For two tied groups of sizes 2 and 3 at distance 4, aggregate coefficient $\lambda = 1.6$ gives objective optima $6.4, 4.8, 3.2$ at $\eta = 0, .25, .5$; $\lambda = 2.4$ gives $8, 6, 4$. At $\lambda = 2$, a continuum of minimizing separations is possible; identical optimizer arrays are therefore not required. The cases $\eta = 0$, $\eta = 1$, $\gamma = 0$, empty/disconnected graphs and duplicate anchors are checked independently.

E2: fixed tangent constraints. Write $\delta_i = g_i - u_i$, $P_i = I - u_i u_i^T$ for nonzero unit anchors. The primal is $J_0(G; U, W_0)$ subject to $u_i^T\delta_i = 0$, $\|\delta_i\| \le r = \tan\theta$ and $\delta_i = 0$ on zero anchors. With $v = B^T q$ its dual is

$$D_T(q) = \langle U, v\rangle - r\sum_{i:u_i\neq 0}(\|P_i v_i\| - 1)_+, \qquad \|q_e\| \le c_e.$$

The proximal displacement projects onto the fixed tangent plane, shrinks radially by the node step and clips to r. The isotropic control omits P_i, uses radius $\sin\theta$ and sums the support correction over all nodes. A consensus outside the constraints cannot bound the primal optimum. Zero rows remain in ACC/NMI; angle masks are diagnostic only.

For the gate, ordinary SSC is selected without labels as the first certified active point in $\{2, 8, 32\}\gamma_{\mathrm{ref}}$. Radial neutralization uses $R_i = u_i + P_i\delta_i$ on nonzero anchors, $R_i = g_i$ otherwise. Energy denominators exclude zero anchors; angles require nonzero anchors and pre-normalization G/R norms above 10^{-12}. An empty or zero-energy diagnostic cannot pass. Two fixed development contrasts use lower quantile .025; both outcomes are retained.

E3: frozen relative-neighbor correction. Two separately centered/normalized feature views use verified checkerboard geometry or the fixed PCG64 feature split. All roles use their common valid

pool and immutable original IDs. Set $q = \min(10, n_v - 1)$ and $Q = \min(4q, n_v - 1)$. Reference j must be mutual-q in both views; contradictory neighbor k is a positive W_0 neighbor outside both first-Q lists. Strict inversion, nonzero separation, bounded initial correction and the registered rank/ID order select at most one triplet per anchor.

With $s = \|u_j - u_k\|$, $m = .1s^2$, $n = (u_k - u_j)/s$ and $\beta = (m + \|u_j\|^2 - \|u_k\|^2)/(2s)$, the normalized hinge is $h(g_i) = [n^T g_i + \beta]_+$. The objective is $J_R = J_0 + \mu \sum h(g_i)$, with no coverage rescaling. For operator T containing those unit rows, the dual is

$$D_R(q, p) = \langle U, B^T q + T^T p \rangle + \beta^T p, \quad \|q_e\| \le c_e, \quad 0 \le p \le \mu, \quad \|(B^T q + T^T p)_i\| \le 1.$$

Steps use the combined $[B; T]$ operator. Feasibility repair scales q, p together. For $\gamma = 0$, $\mu > 1$ gives $g_i = u_i - h(u_i)n_i$; $\mu < 1$ gives U, and $\mu = 1$ permits the intervening segment. With no triplets the target is ordinary SSC and the dataset is still evaluated. Satisfaction is reported separately for G and Q.

Qualification and recovery. Each solver keeps independent feasible primal and dual candidates, including the initial bound. Certificates are recomputed from saved G, q, p, retain signed negative gaps and require relative gap $\le 10^{-5}$ and feasibility/capacity violation $\le 10^{-8}$. Small graph SOCP references use CLARABEL independently of PDHG. Primary continuations have cumulative ceilings 30k/1800, 60k/3600 and 120k/7200 iterations/CPU seconds. Full state and object hashes support continuation. Inactive, uncertified, failed, gated and budget-unrun states are distinct. Extra development sensitivities are $\eta = .5$, $\theta = 30^\circ$ and $\mu = .5$ at the common primary gamma. Matching uses declared certified points only, a 5% tolerance with absolute floors 10^{-6} (normalized L2,1) and $.1^\circ$ (RMS angle), and no added search.

E.1 Registered endpoint status

Data	Study	Panel	Status
ORL	E1	original	certified
ORL	E1	angular	certified
ORL	E2	angular	not promoted
ORL	E3	angular	not promoted
YaleB	E1	original	certified
YaleB	E1	angular	certified
YaleB	E2	angular	not promoted
YaleB	E3	angular	not promoted
AR	E1	original	certified
AR	E1	angular	certified
AR	E2	angular	not promoted
AR	E3	angular	not promoted
JAFFE	E1	original	certified
JAFFE	E1	angular	certified
JAFFE	E2	angular	not promoted
JAFFE	E3	angular	not promoted
UMIST	E1	original	certified
UMIST	E1	angular	certified
UMIST	E2	angular	not promoted
UMIST	E3	angular	not promoted
BinAlpha	E1	original	certified
BinAlpha	E1	angular	certified
BinAlpha	E2	angular	not promoted
BinAlpha	E3	angular	not promoted
MSRA50	E1	original	uncertified
MSRA50	E1	angular	certified
MSRA50	E2	angular	not promoted
MSRA50	E3	angular	not promoted
USPS	E1	original	certified

Data	Study	Panel	Status
USPS	E1	angular	certified
USPS	E2	angular	not promoted
USPS	E3	angular	not promoted

E.2 Complete fixed-point means and numerical qualification

Bold study names denote primary operating points. Means and sample SD use the same complete fifty seeds. Native single-assignment endpoints and all seeds are retained in the CSV/gzip release. Gated or unrun points are absent from this measured table, rather than filled with zeros.

Data	Study	Panel	Parameters	ACC	NMI	Gap	Move	CPU (s)
AR	**SC**	original	gamma=0	57.264±0.833	82.380±0.318	–	0.0000	–
AR	**SSC**	original	gamma=0.2266	57.364±0.857	82.327±0.318	7.63e-06 C	0.0185	3.6
AR	**E1**	original	gamma=0.2266 eta=0.25	57.410±0.758	82.369±0.332	9.73e-06 C	0.0142	7.2
AR	mix	original	gamma=0.2266 eta=0.25	57.483±0.690	82.445±0.317	–	0.0138	–
AR	scale	original	gamma=0 eta=0.25	57.264±0.833	82.380±0.318	–	0.7500	–
AR	**SC**	angular	gamma=0	58.362±0.966	83.232±0.316	–	0.0000	–
AR	**SSC**	angular	gamma=0.1952	58.369±0.769	83.232±0.253	7.04e-06 C	0.0080	3.8
AR	**E1**	angular	gamma=0.1952 eta=0.25	58.317±0.790	83.209±0.228	9.51e-06 C	0.0063	3.9
AR	mix	angular	gamma=0.1952 eta=0.25	58.498±0.776	83.281±0.243	–	0.0060	–
AR	scale	angular	gamma=0 eta=0.25	58.362±0.966	83.232±0.316	–	0.7500	–
JAFFE	**SC**	original	gamma=0	95.305±0.000	94.134±0.020	–	0.0000	–
JAFFE	**SSC**	original	gamma=0.2518	95.305±0.000	94.134±0.020	9.85e-06 C	0.0016	2.2
JAFFE	**E1**	original	gamma=0.2518 eta=0.25	95.305±0.000	94.134±0.020	9.39e-06 C	0.0013	2.6
JAFFE	mix	original	gamma=0.2518 eta=0.25	95.305±0.000	94.134±0.020	–	0.0012	–
JAFFE	scale	original	gamma=0 eta=0.25	95.305±0.000	94.134±0.020	–	0.7500	–
JAFFE	**SC**	angular	gamma=0	96.714±0.000	96.392±0.000	–	0.0000	–
JAFFE	**SSC**	angular	gamma=0.1874	96.714±0.000	96.392±0.000	9.97e-06 C	0.0001	3.2
JAFFE	**E1**	angular	gamma=0.1874 eta=0.25	96.714±0.000	96.392±0.000	9.97e-06 C	0.0001	4.5
JAFFE	mix	angular	gamma=0.1874 eta=0.25	96.714±0.000	96.392±0.000	–	0.0001	–
JAFFE	scale	angular	gamma=0 eta=0.25	96.714±0.000	96.392±0.000	–	0.7500	–
ORL	**SC**	original	gamma=0	78.325±1.310	89.874±0.675	–	0.0000	–
ORL	**SSC**	original	gamma=0.2701	78.405±1.181	89.592±0.677	9.51e-06 C	0.0138	1.7
ORL	**E1**	original	gamma=0.2701 eta=0.25	78.225±1.348	89.798±0.629	9.93e-06 C	0.0105	5.3
ORL	SSC	original	gamma=0	78.325±1.310	89.874±0.675	0 C	0.0000	1.6
ORL	SSC	original	gamma=0.06753	78.325±1.310	89.874±0.675	2e-12 C	0.0000	1.3
ORL	SSC	original	gamma=0.1351	78.325±1.310	89.874±0.675	2e-12 C	0.0000	1.2
ORL	SSC	original	gamma=1	78.480±1.408	89.888±0.805	9.89e-06 C	0.0896	2.1
ORL	SSC	original	gamma=1.08	78.780±1.425	90.140±0.824	9.41e-06 C	0.0928	2.4
ORL	E1	original	gamma=0 eta=0.25	78.325±1.310	89.874±0.675	0 C	0.0000	1.3
ORL	E1	original	gamma=0.06753 eta=0.25	78.325±1.310	89.874±0.675	2e-12 C	0.0000	1.3
ORL	E1	original	gamma=0.1351 eta=0.25	78.325±1.310	89.874±0.675	2e-12 C	0.0000	1.3
ORL	E1	original	gamma=0.2701 eta=0.5	78.030±1.478	89.689±0.698	9.97e-06 C	0.0070	9.7
ORL	E1	original	gamma=1 eta=0.25	78.420±1.358	89.823±0.823	4.86e-05 U	0.0666	26.6
ORL	E1	original	gamma=1.08 eta=0.25	78.450±1.128	89.789±0.675	4.56e-05 U	0.0691	26.3
ORL	mix	original	gamma=0.2701 eta=0.25	78.155±1.257	89.618±0.675	–	0.0104	–
ORL	mix	original	gamma=0.2701 eta=0.5	78.300±1.189	89.696±0.684	–	0.0069	–
ORL	scale	original	gamma=0 eta=0.25	78.325±1.310	89.874±0.675	–	0.7500	–
ORL	**SC**	angular	gamma=0	80.175±1.551	90.890±0.641	–	0.0000	–
ORL	**SSC**	angular	gamma=0.1537	80.180±1.392	90.896±0.610	9.74e-06 C	0.0006	2.2
ORL	**E1**	angular	gamma=0.1537 eta=0.25	80.190±1.361	90.904±0.611	9.64e-06 C	0.0005	2.8
ORL	SSC	angular	gamma=0	80.175±1.551	90.890±0.641	0 C	0.0000	1.3
ORL	SSC	angular	gamma=0.03842	80.175±1.551	90.890±0.641	2e-12 C	0.0000	1.2
ORL	SSC	angular	gamma=0.07683	80.175±1.551	90.890±0.641	2e-12 C	0.0000	1.3
ORL	SSC	angular	gamma=0.6147	80.495±1.394	90.962±0.587	9.9e-06 C	0.0300	5.8
ORL	SSC	angular	gamma=1	80.140±1.147	90.887±0.472	9.96e-06 C	0.0452	7.2
ORL	E1	angular	gamma=0 eta=0.25	80.175±1.551	90.890±0.641	0 C	0.0000	1.3

Data	Study	Panel	Parameters	ACC	NMI	Gap	Move	CPU (s)
ORL	E1	angular	gamma=0.03842 eta=0.25	80.175±1.551	90.890±0.641	2e-12 C	0.0000	1.2
ORL	E1	angular	gamma=0.07683 eta=0.25	80.175±1.551	90.890±0.641	2e-12 C	0.0000	1.3
ORL	E1	angular	gamma=0.1537 eta=0.5	80.185±1.364	90.888±0.597	9.86e-06 C	0.0004	5.8
ORL	E1	angular	gamma=0.6147 eta=0.25	80.350±1.415	90.949±0.594	2.02e-05 U	0.0225	31.0
ORL	E1	angular	gamma=1 eta=0.25	80.485±1.330	91.060±0.498	4.42e-05 U	0.0335	31.1
ORL	mix	angular	gamma=0.1537 eta=0.25	80.255±1.356	90.923±0.606	–	0.0004	–
ORL	mix	angular	gamma=0.1537 eta=0.5	80.165±1.394	90.888±0.615	–	0.0003	–
ORL	scale	angular	gamma=0 eta=0.25	80.175±1.551	90.890±0.641	–	0.7500	–
ORL	radial R	angular	gamma=0.1537	80.180±1.392	90.896±0.610	–	0.0006	–
UMIST	**SC**	original	gamma=0	61.901±2.505	79.376±1.121	–	0.0000	–
UMIST	**SSC**	original	gamma=0.2927	61.890±2.251	79.122±1.201	5.81e-06 C	0.0047	5.3
UMIST	**E1**	original	gamma=0.2927 eta=0.25	62.528±2.087	79.235±1.028	9.81e-06 C	0.0037	30.2
UMIST	mix	original	gamma=0.2927 eta=0.25	62.103±2.120	79.157±1.088	–	0.0035	–
UMIST	scale	original	gamma=0 eta=0.25	61.901±2.505	79.376±1.121	–	0.7500	–
UMIST	**SC**	angular	gamma=0	66.638±2.495	83.897±0.756	–	0.0000	–
UMIST	**SSC**	angular	gamma=0.1947	66.539±2.462	83.871±0.726	9.98e-06 C	0.0014	4.3
UMIST	**E1**	angular	gamma=0.1947 eta=0.25	66.543±2.499	83.868±0.760	9.99e-06 C	0.0011	8.1
UMIST	mix	angular	gamma=0.1947 eta=0.25	66.738±2.373	83.936±0.735	–	0.0010	–
UMIST	scale	angular	gamma=0 eta=0.25	66.638±2.495	83.897±0.756	–	0.7500	–
BinAlpha	**SC**	original	gamma=0	47.272±1.371	58.664±0.582	–	0.0000	–
BinAlpha	**SSC**	original	gamma=0.2931	47.387±1.263	58.530±0.762	9.66e-06 C	0.0269	4.3
BinAlpha	**E1**	original	gamma=0.2931 eta=0.25	47.803±1.304	58.822±0.661	9.83e-06 C	0.0222	7.4
BinAlpha	mix	original	gamma=0.2931 eta=0.25	47.313±1.256	58.498±0.610	–	0.0202	–
BinAlpha	scale	original	gamma=0 eta=0.25	47.272±1.371	58.664±0.582	–	0.7500	–
BinAlpha	**SC**	angular	gamma=0	50.736±1.411	61.005±0.631	–	0.0000	–
BinAlpha	**SSC**	angular	gamma=0.2379	50.879±1.305	61.104±0.621	8.76e-06 C	0.0126	3.3
BinAlpha	**E1**	angular	gamma=0.2379 eta=0.25	51.128±1.378	61.204±0.522	7.65e-06 C	0.0113	2.5
BinAlpha	mix	angular	gamma=0.2379 eta=0.25	51.064±1.456	61.014±0.652	–	0.0094	–
BinAlpha	scale	angular	gamma=0 eta=0.25	50.736±1.411	61.005±0.631	–	0.7500	–
MSRA50	**SC**	original	gamma=0	47.117±2.479	60.281±2.456	–	0.0000	–
MSRA50	**SSC**	original	gamma=0.2133	46.976±2.295	60.193±2.273	1.01e-05 U	0.0002	179.3
MSRA50	**E1**	original	gamma=0.2133 eta=0.25	46.966±2.283	60.259±2.327	1.09e-05 U	0.0002	242.6
MSRA50	mix	original	gamma=0.2133 eta=0.25	46.966±2.283	60.259±2.327	–	0.0002	–
MSRA50	scale	original	gamma=0 eta=0.25	47.117±2.479	60.281±2.456	–	0.7500	–
MSRA50	**SC**	angular	gamma=0	57.434±2.584	72.643±1.027	–	0.0000	–
MSRA50	**SSC**	angular	gamma=0.1275	57.435±2.585	72.647±1.029	9.99e-06 C	0.0000	73.2
MSRA50	**E1**	angular	gamma=0.1275 eta=0.25	57.364±2.589	72.632±1.029	9.99e-06 C	0.0000	88.9
MSRA50	mix	angular	gamma=0.1275 eta=0.25	57.363±2.588	72.628±1.027	–	0.0000	–
MSRA50	scale	angular	gamma=0 eta=0.25	57.434±2.584	72.643±1.027	–	0.7500	–
YaleB	**SC**	original	gamma=0	21.781±0.454	27.731±0.585	–	0.0000	–
YaleB	**SSC**	original	gamma=0.2485	21.789±0.470	27.681±0.683	9.86e-06 C	0.0049	8.5
YaleB	**E1**	original	gamma=0.2485 eta=0.25	21.819±0.465	27.683±0.552	1e-05 C	0.0045	25.1
YaleB	SSC	original	gamma=0	21.781±0.454	27.731±0.585	0 C	0.0000	2.0
YaleB	SSC	original	gamma=0.06212	21.781±0.454	27.731±0.585	2e-12 C	0.0000	2.0
YaleB	SSC	original	gamma=0.1242	21.781±0.454	27.731±0.585	2e-12 C	0.0000	2.1
YaleB	SSC	original	gamma=0.9939	21.586±0.497	27.539±0.575	9.98e-06 C	0.0971	93.3
YaleB	SSC	original	gamma=1	21.511±0.484	27.455±0.576	9.98e-06 C	0.0976	77.8
YaleB	E1	original	gamma=0 eta=0.25	21.781±0.454	27.731±0.585	0 C	0.0000	2.0
YaleB	E1	original	gamma=0.06212 eta=0.25	21.781±0.454	27.731±0.585	2e-12 C	0.0000	2.3
YaleB	E1	original	gamma=0.1242 eta=0.25	21.781±0.454	27.731±0.585	2e-12 C	0.0000	2.1
YaleB	E1	original	gamma=0.2485 eta=0.5	21.811±0.492	27.736±0.575	9.9e-06 C	0.0038	40.4
YaleB	E1	original	gamma=0.9939 eta=0.25	21.678±0.428	27.433±0.507	5.3e-05 U	0.0746	157.2
YaleB	E1	original	gamma=1 eta=0.25	21.708±0.473	27.562±0.566	5.33e-05 U	0.0749	155.3
YaleB	mix	original	gamma=0.2485 eta=0.25	21.752±0.548	27.687±0.528	–	0.0036	–

Data	Study	Panel	Parameters	ACC	NMI	Gap	Move	CPU (s)
YaleB	mix	original	gamma=0.2485 eta=0.5	21.857±0.472	27.758±0.502	–	0.0024	–
YaleB	scale	original	gamma=0 eta=0.25	21.781±0.454	27.731±0.585	–	0.7500	–
YaleB	**SC**	angular	gamma=0	25.797±0.709	32.595±0.435	–	0.0000	–
YaleB	**SSC**	angular	gamma=0.2068	26.198±0.953	32.743±0.649	8.5e-06 C	0.0043	3.0
YaleB	**E1**	angular	gamma=0.2068 eta=0.25	25.984±0.885	32.635±0.576	8.71e-06 C	0.0040	3.8
YaleB	SSC	angular	gamma=0	25.797±0.709	32.595±0.435	0 C	0.0000	1.4
YaleB	SSC	angular	gamma=0.0517	25.797±0.709	32.595±0.435	2e-12 C	0.0000	1.5
YaleB	SSC	angular	gamma=0.1034	25.797±0.709	32.595±0.435	2e-12 C	0.0000	1.5
YaleB	SSC	angular	gamma=0.8273	25.570±0.798	32.232±0.472	9.63e-06 C	0.1000	11.3
YaleB	SSC	angular	gamma=1	25.356±0.780	31.931±0.512	9.75e-06 C	0.1171	9.2
YaleB	E1	angular	gamma=0 eta=0.25	25.797±0.709	32.595±0.435	0 C	0.0000	1.7
YaleB	E1	angular	gamma=0.0517 eta=0.25	25.797±0.709	32.595±0.435	2e-12 C	0.0000	1.5
YaleB	E1	angular	gamma=0.1034 eta=0.25	25.797±0.709	32.595±0.435	2e-12 C	0.0000	1.7
YaleB	E1	angular	gamma=0.2068 eta=0.5	25.884±0.945	32.750±0.536	9.82e-06 C	0.0034	4.2
YaleB	E1	angular	gamma=0.8273 eta=0.25	25.537±0.730	32.107±0.431	9.89e-06 C	0.0779	35.9
YaleB	E1	angular	gamma=1 eta=0.25	25.366±0.812	32.029±0.543	9.97e-06 C	0.0901	40.4
YaleB	mix	angular	gamma=0.2068 eta=0.25	25.940±0.787	32.674±0.505	–	0.0032	–
YaleB	mix	angular	gamma=0.2068 eta=0.5	25.780±0.774	32.638±0.495	–	0.0021	–
YaleB	scale	angular	gamma=0 eta=0.25	25.797±0.709	32.595±0.435	–	0.7500	–
YaleB	radial R	angular	gamma=0.2068	26.145±0.983	32.672±0.639	–	0.0042	–
USPS	**SC**	original	gamma=0	65.597±1.867	80.315±2.128	–	0.0000	–
USPS	**SSC**	original	gamma=0.2496	65.550±1.680	80.419±1.724	9.97e-06 C	0.0008	150.0
USPS	**E1**	original	gamma=0.2496 eta=0.25	65.604±1.633	80.572±1.396	9.99e-06 C	0.0008	330.9
USPS	mix	original	gamma=0.2496 eta=0.25	65.664±1.686	80.502±1.459	–	0.0006	–
USPS	scale	original	gamma=0 eta=0.25	65.597±1.867	80.315±2.128	–	0.7500	–
USPS	**SC**	angular	gamma=0	68.360±3.635	83.436±0.891	–	0.0000	–
USPS	**SSC**	angular	gamma=0.189	68.124±3.866	83.322±1.130	9.32e-06 C	0.0004	8.3
USPS	**E1**	angular	gamma=0.189 eta=0.25	68.136±3.462	83.430±0.889	9.74e-06 C	0.0003	12.3
USPS	mix	angular	gamma=0.189 eta=0.25	68.124±3.867	83.322±1.130	–	0.0003	–
USPS	scale	angular	gamma=0 eta=0.25	68.360±3.635	83.436±0.891	–	0.7500	–

E.3 All declared paired contrasts

Intervals use 10,000 paired resamples of rounding seeds, conditional on the fixed input and saved representation. Qualification requires both applicable convex certificates. Interpolation, radial neutralization and SC are deterministic references, not certified optimizers of another target.

Data	Panel	Contrast	ΔACC [interval]	ΔNMI	Qualified
ORL	original	E1_minus_SC	-0.100 [-0.575, 0.385]	-0.076	True
ORL	original	E1_minus_SSC	-0.180 [-0.660, 0.315]	0.206	True
ORL	original	E1_minus_interpolation	0.070 [-0.295, 0.405]	0.180	True
ORL	original	E1_minus_uniform_scaling	-0.100 [-0.575, 0.385]	-0.076	True
ORL	angular	E1_minus_SC	0.015 [-0.150, 0.225]	0.014	True
ORL	angular	E1_minus_SSC	0.010 [-0.015, 0.045]	0.008	True
ORL	angular	E1_minus_interpolation	-0.065 [-0.175, 0.000]	-0.019	True
ORL	angular	E1_minus_uniform_scaling	0.015 [-0.150, 0.225]	0.014	True
YaleB	original	E1_minus_SC	0.037 [-0.155, 0.234]	-0.048	True
YaleB	original	E1_minus_SSC	0.030 [-0.103, 0.169]	0.002	True
YaleB	original	E1_minus_interpolation	0.066 [-0.067, 0.195]	-0.004	True
YaleB	original	E1_minus_uniform_scaling	0.037 [-0.155, 0.234]	-0.048	True
YaleB	angular	E1_minus_SC	0.187 [-0.151, 0.516]	0.041	True
YaleB	angular	E1_minus_SSC	-0.214 [-0.412, -0.034]	-0.108	True
YaleB	angular	E1_minus_interpolation	0.045 [-0.152, 0.247]	-0.038	True
YaleB	angular	E1_minus_uniform_scaling	0.187 [-0.151, 0.516]	0.041	True
AR	original	E1_minus_SC	0.145 [-0.131, 0.438]	-0.011	True
AR	original	E1_minus_SSC	0.045 [-0.233, 0.333]	0.042	True
AR	original	E1_minus_interpolation	-0.074 [-0.317, 0.183]	-0.076	True
AR	original	E1_minus_uniform_scaling	0.145 [-0.131, 0.438]	-0.011	True
AR	angular	E1_minus_SC	-0.045 [-0.400, 0.310]	-0.023	True
AR	angular	E1_minus_SSC	-0.052 [-0.331, 0.233]	-0.022	True
AR	angular	E1_minus_interpolation	-0.181 [-0.429, 0.067]	-0.072	True
AR	angular	E1_minus_uniform_scaling	-0.045 [-0.400, 0.310]	-0.023	True

Data	Panel	Contrast	ΔACC [interval]	ΔNMI	Qualified
JAFFE	original	E1_minus_SC	0.000 [0.000, 0.000]	-0.000	True
JAFFE	original	E1_minus_SSC	0.000 [0.000, 0.000]	0.000	True
JAFFE	original	E1_minus_interpolation	0.000 [0.000, 0.000]	0.000	True
JAFFE	original	E1_minus_uniform_scaling	0.000 [0.000, 0.000]	-0.000	True
JAFFE	angular	E1_minus_SC	0.000 [0.000, 0.000]	0.000	True
JAFFE	angular	E1_minus_SSC	0.000 [0.000, 0.000]	0.000	True
JAFFE	angular	E1_minus_interpolation	0.000 [0.000, 0.000]	0.000	True
JAFFE	angular	E1_minus_uniform_scaling	0.000 [0.000, 0.000]	0.000	True
UMIST	original	E1_minus_SC	0.628 [-0.096, 1.344]	-0.141	True
UMIST	original	E1_minus_SSC	0.638 [0.082, 1.245]	0.113	True
UMIST	original	E1_minus_interpolation	0.426 [0.067, 0.862]	0.077	True
UMIST	original	E1_minus_uniform_scaling	0.628 [-0.096, 1.344]	-0.141	True
UMIST	angular	E1_minus_SC	-0.096 [-0.461, 0.206]	-0.028	True
UMIST	angular	E1_minus_SSC	0.004 [-0.238, 0.241]	-0.003	True
UMIST	angular	E1_minus_interpolation	-0.195 [-0.504, 0.000]	-0.067	True
UMIST	angular	E1_minus_uniform_scaling	-0.096 [-0.461, 0.206]	-0.028	True
BinAlpha	original	E1_minus_SC	0.531 [0.021, 1.039]	0.158	True
BinAlpha	original	E1_minus_SSC	0.415 [-0.049, 0.885]	0.292	True
BinAlpha	original	E1_minus_interpolation	0.490 [0.069, 0.931]	0.324	True
BinAlpha	original	E1_minus_uniform_scaling	0.531 [0.021, 1.039]	0.158	True
BinAlpha	angular	E1_minus_SC	0.392 [-0.200, 0.972]	0.199	True
BinAlpha	angular	E1_minus_SSC	0.249 [-0.087, 0.572]	0.101	True
BinAlpha	angular	E1_minus_interpolation	0.064 [-0.336, 0.451]	0.190	True
BinAlpha	angular	E1_minus_uniform_scaling	0.392 [-0.200, 0.972]	0.199	True
MSRA50	original	E1_minus_SC	-0.151 [-0.434, 0.008]	-0.022	False
MSRA50	original	E1_minus_SSC	-0.010 [-0.030, 0.000]	0.066	False
MSRA50	original	E1_minus_interpolation	0.000 [0.000, 0.000]	0.000	False
MSRA50	original	E1_minus_uniform_scaling	-0.151 [-0.434, 0.008]	-0.022	False
MSRA50	angular	E1_minus_SC	-0.070 [-0.213, 0.003]	-0.012	True
MSRA50	angular	E1_minus_SSC	-0.071 [-0.213, 0.000]	-0.016	True
MSRA50	angular	E1_minus_interpolation	0.001 [0.000, 0.003]	0.004	True
MSRA50	angular	E1_minus_uniform_scaling	-0.070 [-0.213, 0.003]	-0.012	True
USPS	original	E1_minus_SC	0.006 [-0.428, 0.405]	0.258	True
USPS	original	E1_minus_SSC	0.053 [-0.003, 0.163]	0.153	True
USPS	original	E1_minus_interpolation	-0.060 [-0.184, 0.003]	0.070	True
USPS	original	E1_minus_uniform_scaling	0.006 [-0.428, 0.405]	0.258	True
USPS	angular	E1_minus_SC	-0.224 [-0.686, 0.159]	-0.006	True
USPS	angular	E1_minus_SSC	0.012 [-0.490, 0.526]	0.108	True
USPS	angular	E1_minus_interpolation	0.011 [-0.491, 0.525]	0.108	True
USPS	angular	E1_minus_uniform_scaling	-0.224 [-0.686, 0.159]	-0.006	True

E.4 Development mechanism decisions

E2: not promoted. Both predefined development outcomes follow. ORL: FAILED. Inward energy share 0.001, RMS angle 0.000 degrees. ACC change 0.000 pp, adjusted lower 0.000 pp; NMI change 0.000 pp.

YaleB: FAILED. Inward energy share 0.055, RMS angle 0.018 degrees. ACC change -0.053 pp, adjusted lower -0.183 pp; NMI change -0.072 pp.

E3: not promoted. Both predefined development outcomes follow. ORL: FAILED. Comparisons 49 / required 20; correct fraction 0.224, reversed fraction 0.082.

YaleB: FAILED. Comparisons 1751 / required 121; correct fraction 0.448, reversed fraction 0.019.

Interpretation. Dispersion and hinge diagnostics concern the saved optimized G. Output normalization may alter hinge satisfaction and angular movement. Numerical activity, retained dispersion and passing a development gate do not by themselves establish ACC/NMI gains beyond the named controls. Full saved-array diagnostics, conditions, masks, failures and object identities accompany the scalar release.